

Six Sheets, Three Escapes

Complete Fiber Geometry and Explicit Normalization
of a Five-Dimensional Keller Map

A research extension of the Jacobian-conjecture
direction in the ProveIt repository

Merged report of two manuscripts: *Six Sheets, Three Escapes* (source 02, the base) and *A Codimension-Three Omitted Surface for a Five-Dimensional Keller Map* (source 05)

Both sources dated 24 September 2026

Principal result. We determine every geometric fiber of Gao's map F_6 , including the previously untreated degeneration at its first target coordinate zero. We construct its finite normalization with an explicit six-element free basis, identify its omitted surface and its two-component nonproperness locus, and prove geometric and arithmetic S_6 results. Source 05 adds every root-multiplicity pattern, every local analytic type of the discriminant, normal crossings with local monodromy $(\mathbb{Z}/2)^3$ along the omitted surface, an obstruction to stabilization from dimension at most three, and a second arithmetic S_6 certificate.

Status. The map F_6 , its polynomiality and its generic degree are Gao's prior work. The extensions are proved here with supplementary exact computer-algebra checks. Both sources are AI-assisted research drafts prepared with ChatGPT; neither is a peer-reviewed paper or a Lean/Rocq-certified development, and nothing about F_6 is formalized in ProveIt. Priority beyond the inspected sources has not been established.

Prepared for the ProveIt research program. Both sources' reproducibility scripts, recorded outputs and sparse coefficient data accompany this report (Appendix A).

Abstract

Gao's five-dimensional polynomial map F_6 has constant Jacobian determinant -290 and generic geometric degree six. Its published construction provides a sextic controlling fibers away from a degenerating target hyperplane, but leaves the complete fiber stratification to further work. We resolve the fiber-cardinality problem for this particular map and give additional structural results.

The decisive identity is that the sweep parameter is the derivative of the fiber sextic on its zero set. After a rescaling, this produces a uniform root chart valid even on the exceptional hyperplane: the source open set $x \neq 0$ is exactly a simple-root locus. The remaining source hyperplane maps isomorphically to the target hyperplane. Consequently the only geometric fiber cardinalities are $0, 1, 2, 3, 4, 6$, all of which occur. The omitted set is an explicitly parametrized smooth closed surface isomorphic to $\mathbb{G}_m \times \mathbb{A}^1$.

We then normalize the monogenic sextic order by adjoining one explicit fraction. The resulting algebra is free of rank six, with three short multiplication rules, and its spectrum is smooth. The original affine space is the complement of the ramification locus and three unramified boundary components in this finite model. Its nonproperness locus is $V(cH)$, where H is an irreducible normalized discriminant satisfying $H|_{c=0} = -108(A^2 + 4B)$. A one-parameter Morse slice proves geometric monodromy S_6 ; reductions modulo $3, 13, 37$ certify an explicit rational specialization with the same Galois group. We also determine the possible real fiber sizes, extract a general normalization lemma, and give an exact inversion procedure and a proposed formalization and research agenda.

The second source, merged here, proves the fiber-size classification independently and adds finer structure. Newton sums show that every sextic of the family has at least three distinct roots, so exactly seven root-multiplicity patterns can occur, and all seven do. The discriminant surface in the three-dimensional coefficient space is normalized by the affine plane, and its local analytic type is determined at every point. At each point of the omitted surface the discriminant component of the nonproperness locus has three smooth, mutually transverse branches, with local monodromy $(\mathbb{Z}/2)^3$. The omitted surface excludes stabilization from any polynomial self-map in at most three variables. The generic inverse field extension has no proper intermediate field. A second Morse slice reproves geometric S_6 , and reductions modulo $7, 11, 269$ certify a second rational specialization with Galois group S_6 .

Keywords. Keller map; polynomial fiber; normalization; nonproperness; discriminant; monodromy; sextic; radical obstruction.

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1 The research target and the main conclusions

1.1 Relation to ProveIt and to the published problem

The Jacobian-conjecture project in ProveIt contains explicit polynomial maps, exact collision certificates, symmetry arguments, and stable reductions of degree. Its research README studies degree and sparsity questions for the three-dimensional example. The repository walkthrough also points to Gao’s account of the subsequent higher-dimensional constructions [1, 3]. These give a natural opportunity to pass from individual collision witnesses to the full geometry of an explicit map.

We focus on the map denoted F_6 in [3, §4.5.1, Theorem 4.5]. The subscript labels the example; its *ambient dimension is five*. The map, its determinant, its sextic elimination equation, and the four-point fibers over its nonzero distinguished axis are prior results. Gao’s concluding discussion and Appendix A.3 explicitly leave the complete higher-dimensional stratifications, including that of F_6 , unresolved in version 1. We do not claim to have constructed a new counterexample to the Jacobian conjecture. Our object is the full fiber geometry and finite algebraic model of this existing example.

The source audit was scoped, not repository-wide. It covered the root README, the Jacobian project and its research README, the relevant walkthrough search result, and the cited primary paper. The exact repository revision and inspected file hashes are recorded in Appendix B. No claim is made that every research artifact in this large repository has been searched for equivalent results.

Source 05’s framing, kept. Source 05 describes the same context: the **Algebra/JacobianConjecture** development contains an explicit polynomial counterexample, determinant and collision identities, and research on stable reductions and equivariance, and its walkthrough points to Gao’s broader tangent-sweep construction as external mathematical context [1, 2]. (“Tangent-sweep” is source 05’s gloss; the walkthrough does not use the word.) Source 05 states explicitly that repository theorem names are not to be confused with theorems established here. It also records what it does not claim: F_6 is not a newly discovered construction; neither the plane Jacobian problem nor Gao’s separate middle-branch rigidity problem is settled; and no complete Whitney stratification or irreducible decomposition of every higher-contact incidence variety is claimed. Those are distinct, finer questions.

1.2 This merged report and ProveIt’s formal development

This report is built from two manuscripts of ProveIt’s incoming batch 36, both pinned to ProveIt commit `e21766d04` and both dated 24 September 2026. Source 02, *Six Sheets, Three Escapes*, is the base; its text is kept, in its order, as Sections 1–12 and Appendices A–B. Source 05, *A Codimension-Three Omitted Surface for a Five-Dimensional Keller Map*, proves the same spine (derivative identity, simple-root reconstruction, spectrum, omitted surface, nonproperness, S_6) in different notation, and adds results source 02 does not have. Every result, proof, example, limitation and question of source 05 is printed here, in source 02’s notation (Section 1.4). A result proved by both is printed once and credited to both; where source 05’s proof is genuinely different it is kept as a marked second route. Statements, proofs and remarks taken from source 05 carry the tag (source 05) in their heading or first sentence. Untagged text is source 02’s, except the editorial text written for this merge: Sections 1.2 and 1.4, the rewritten Appendix A, Appendix D, Remark 9.2, the re-scoping notes in Section 11, and short linking sentences that say which source proved what. Appendix D records the provenance and every choice the merge made.

Formal status. Nothing in this report is formalized. In particular, the ProveIt Lean and Rocq developments contain no statement about F_6 . Their five-variable declarations, such as

`jacobianConjectureInDimensionFive_false` (Algebra/JacobianConjecture/Lean/JacobianConjecture/SimplerCounterexample.lean) and `jacobian_conjecture_dimension_five_is_false` (Algebra/JacobianConjecture/Coq/SimplerCounterexample.v), concern the project's own stable map $G = (G_1, \dots, G_5)$: a stable shear of Alpöge's three-dimensional map, of ordinary degree six with profile $(6, 6, 4, 3, 4)$ and a two-point collision. That G is *not* F_6 . The map F_6 is Gao's [3]; it has ambient dimension five, geometric degree six, and ordinary coordinate degrees $(7, 38, 40, 42, 44)$. This report belongs to ProveIt's research-report collection and continues the formal project `Algebra/JacobianConjecture`; that relation confers no formal status on it. Source 05 says the same of its own assertions: they have not been formalized in Lean or Rocq, have not undergone external peer review, and do not inherit machine-checked status from related ProveIt files; and its comparison with the pinned repository and Gao's version 1 is not an exhaustive priority certification.

1.3 Conventions

Unless indicated otherwise, k is a field of characteristic zero. Equations and chart isomorphisms are defined over $\mathbb{Q}$ and extend to k . A *geometric fiber* means the fiber after passage to an algebraic closure of its residue field, and its cardinality counts distinct points, not multiplicities. Statements about complex nonproperness and monodromy use $k = \mathbb{C}$. Real and rational fibers are treated separately.

Write a source point as (x, y, z_1, z_2, z_3) and a target point as

$$b = (c, A, B, D, E).$$

The use of D rather than C for the fourth coordinate avoids confusion with the field $\mathbb{C}$. Put

$$\kappa = E + B^2 - AD, \quad T = -A + 2cB - c^3\kappa, \quad L = B - cD. \quad (1)$$

The coordinate change from E to κ is a polynomial automorphism of target affine space.

Source 05 states its algebraic assertions about geometric fibers over an algebraically closed field k of characteristic zero and its analytic assertions over $\mathbb{C}$; all its displayed formulas are defined over $\mathbb{Q}$. This is the geometric-fiber setting above, so its proofs need no re-reading for stronger hypotheses. It also fixes three words used throughout. A *Keller map* is a polynomial map with nonzero constant Jacobian determinant. A *simple root* has multiplicity exactly one; a repeated root contributes *no* point to the corresponding fiber, rather than one point. The *omitted locus* is the complement of the image; it can be strictly smaller than the nonproperness locus (Definition before Theorem 6.4).

1.4 Notation, and the dictionary between the two sources

Both sources use identical normalizations up to renaming; the merge checked the dictionary below symbolically. This report uses source 02's names throughout. Source 05's formulas are printed after the renaming in Table 1; no normalization or scalar changes.

Tempting false readings. Several letters mean different things in the two sources, and some also clash with ProveIt's project notation.

- c is the *first target coordinate* here; source 05's c is the constant coefficient of the sextic, here d . Likewise source 05's b, a are coefficients, while here b is also the target point (context decides).
- T here is the target function $-A + 2cB - c^3\kappa$ of (1), never a root variable; source 05's root variable T is w here.

Object	This report (= source 02)	Source 05
target point	$b = (c, A, B, D, E)$	$q = (C, p_1, p_2, p_3, p_4)$
scaled target	$Y_i = c^i \cdot (A, B, D, E)_i$	$Y_i = C^i p_i$
stage monomials	$a_1, \dots, a_4$	$\xi_1, \dots, \xi_4$
sweep variables	(w_1, w_2, w_3)	(t, s, r)
potentials	G_2, G_3, G_4	$A, 2tA + s, B$
sweep quotients S_i/γ^i	$\widehat{S}_i(\gamma, u_1, u_2, u_3)$	E_i
section coordinates at $x = 0$	$\beta_1, \dots, \beta_4$, right side of (15)	jets $J_1, \dots, J_4$
fiber sextic, root	$P_b(w), w = \gamma u_1$	$R_{a,b,c}(T), T = \gamma u_1$
sextic coefficients	(a, b, d) of (51)	(a, b, c)
universal discriminant	$\Delta(a, b, d)$	$D(a, b, c)$
discriminant surface	$\mathcal{D} = V(\Delta) \subset \mathbb{A}_{a,b,d}^3$	$\mathcal{D} = V(D)$
normalized discriminant	$H = \Delta/c^2$	$\widetilde{D}$
hyperplane quadratic's discriminant	$A^2 + 4B$	$H := p_1^2 + 4p_2$
free parameter on M	$t = c^2 B$	$\eta = Y_2$
omitted surface	M	$\mathcal{M}$
nonproperness set	$\text{NP}(\mathcal{F})$	S_F
the cubic with $P = 2Q_\star^2$	$Q_\star(w)$	$Q(T)$
triple-root cofactor; factors	$Q_\rho; h_4(\rho), \Lambda(\rho)$	$Q_\rho; H_4(\rho), K(\rho)$
power sums of the roots	N_1, N_2, N_3	p_1, p_2, p_3 (temporary)
two-root gap $\alpha - \beta$	θ	δ
local analytic coordinates	$(\sigma_1, \sigma_2, \sigma_3)$, quartic variable ω	$(u, v, w), Z$
second Morse slice	$\hat{p}(w) + \tau$	$P(T) + c$
second arithmetic target	$b_{**} = (1, -1, 0, 1, 1)$	q_*

Table 1: Dictionary. Explicitly, $a = 1 + cA = Y_1 + 1$, $b = c^4\kappa - 2c^2B + cA = Y_4 + Y_2^2 - Y_1Y_3 - 2Y_2 + Y_1$ and $d = c^3D - c^2B = Y_3 - Y_2$.

- H here is the normalized discriminant (source 05's $\tilde{D}$), not source 05's $H = p_1^2 + 4p_2$, which is written $A^2 + 4B$ here.
- A, B here are target coordinates, not source 05's sweep potentials; D, E are target coordinates, not a discriminant and not quotient polynomials. E_1, E_2, E_3 are the three escape components of Theorem 5.2.
- $J(w) = g(w) + cA$ is a polynomial in w here, not a boundary jet.
- $\mathcal{N} = \mathcal{N}$ denotes *only* the normalization algebra of F_6 . Source 02 also wrote $\mathcal{N}_{\mathcal{F}}$ for the nonproperness set; this report renames that set $\text{NP}(\mathcal{F})$ (merge disclosure).
- “Normalization” has two objects: the normalization $Z = \text{Spec } \mathcal{N}$ of the map F_6 (Section 5), and the normalization $\mathbb{A}_{a,\rho}^2 \rightarrow \mathcal{D}$ of the discriminant *surface* in coefficient space (Proposition 6.2). “Normalized discriminant” H refers to the first.
- t is the parameter on M in (42) and the Morse parameter in Section 8; source 05's sweep parameter t is w_1 here.
- Project notation: ProveIt's $F = (P, Q, R)$ is Alpöge's three-dimensional map, and its $G = (G_1, \dots, G_5)$ is the five-variable stable map named above. Neither is F_6 ; here P_b, q_b, G_2, G_3, G_4 are unrelated to them. “Degree six” for F_6 means generic fiber size six (geometric degree), not the ordinary degree of the project's G .

1.5 A precise statement of what is proved

Define

$$g(w) = 2w^4 - 2w^3 - w + 1 = (w - 1)(2w^3 - 1), \quad (2)$$

$$J(w) = g(w) + cA, \quad (3)$$

$$P_b(w) = w^2 J(w) - cTw - c^2 L, \quad (4)$$

$$q_b(v) = v^2 J(cv) - Tv - L. \quad (5)$$

Thus $P_b(cv) = c^2 q_b(v)$. Expanded, the uniform polynomial is

$$\begin{aligned} q_b(v) &= 2c^4 v^6 - 2c^3 v^5 - cv^3 + (1 + cA)v^2 \\ &\quad + (c^3 \kappa - 2cB + A)v + cD - B. \end{aligned} \quad (6)$$

Theorem 1.1 (Main conclusions). *For Gao's map $\mathcal{F} = F_6$ the following hold.*

- (i) *The source open set $x \neq 0$ is isomorphic over target space to*

$$\{(b, v) : q_b(v) = 0, \ q'_b(v) \neq 0\}.$$

The source hyperplane $x = 0$ maps isomorphically onto $c = 0$.

- (ii) *Every geometric fiber has*

$$\#\mathcal{F}^{-1}(b) = \#\{\text{simple roots of } q_b\} + \mathbf{1}_{c=0}.$$

For $c = 0$ this equals 3 when $A^2 + 4B \neq 0$ and 1 otherwise. The complete set of attained cardinalities is $\{0, 1, 2, 3, 4, 6\}$.

- (iii) *The omitted set over an algebraically closed field is the smooth closed surface $M \simeq \mathbb{G}_m \times \mathbb{A}^1$ defined by*

$$\begin{aligned} cA &= -\frac{11}{32}, \\ c^3D - c^2B &= \frac{25}{128}, \\ c^4\kappa - 2c^2B + cA &= \frac{5}{32}. \end{aligned} \tag{7}$$

In particular, $\mathcal{F}(\mathbb{A}^5) = \mathbb{A}^5 \setminus M$ on geometric points.

- (iv) *The integral closure of the target coordinate ring in the source function field is free of rank six with basis*

$$1, w, w^2, w^3, w^4, \zeta.$$

Its spectrum is smooth. Its multiplication is determined by

$$\begin{aligned} 2w^5 &= c\zeta + 2w^4 + w^2 - (1 + cA)w, \\ w\zeta &= Tw + cL, \\ \zeta^2 &= T\zeta + LJ(w). \end{aligned} \tag{8}$$

- (v) *Let $H = \text{Disc}_w(P_b)/c^2$. This is an irreducible polynomial, and*

$$H(0, A, B, D, E) = -108(A^2 + 4B).$$

The finite model's discriminant in the displayed basis is $H/256$. The complex nonproperness locus of $\mathcal{F}$ is precisely $V(cH)$.

- (vi) *The geometric monodromy of the six-sheeted cover is S_6 . The rational target $(1, 0, 0, 1, 0)$ gives an irreducible sextic with Galois group S_6 over $\mathbb{Q}$, so a full inverse vector there cannot be expressed using radicals over $\mathbb{Q}$.*

The theorem will be proved in components. In particular, the displayed sextic is *not* itself the finite normalization: at $c = 0$ it collapses two branches to $w = 0$. Ignoring this distinction gives wrong exceptional fiber counts and an artificial discriminant component.

Source 05 states its main results as the following two theorems, printed here in this report's notation. Parts (ii) and (iii) of Theorem 1.2 and the first two sentences of Theorem 1.3 repeat parts (ii), (iii), (v) and (vi) of Theorem 1.1; both sources prove them.

Theorem 1.2 (Complete fibers and image (source 05)). *For $\mathcal{F} = F_6$ the following hold.*

- (i) *Every fiber is finite and reduced. If $c \neq 0$, its points are in explicit bijection with the simple roots of P_b . The only possible root-multiplicity patterns of P_b , and the corresponding fiber sizes, are*

<i>pattern</i>	1^6	$2\,1^4$	$3\,1^3$	$2^2\,1^2$	$4\,1^2$	$3\,2\,1$	2^3
$\#\mathcal{F}^{-1}(b)$	6	4	3	2	2	1	0

and each pattern occurs.

- (ii) *If $c = 0$, then $\#\mathcal{F}^{-1}(b) = 3$ when $A^2 + 4B \neq 0$ and 1 when $A^2 + 4B = 0$. The set of all fiber sizes is exactly $\{0, 1, 2, 3, 4, 6\}$.*
- (iii) *The complement of the image is exactly the surface M of (42). It is a smooth closed complete intersection in $\mathbb{A}^5$, isomorphic to $\mathbb{G}_m \times \mathbb{A}^1$, of codimension three.*
- (iv) *The map is not polynomial-equivalent to a stabilization of any polynomial self-map in at most three variables.*

The geometry is stronger than a list of sample collisions (source 05): the theorem identifies every target with no preimage, every possible loss of sheets, and an exact algebraic decision rule for arbitrary targets.

Theorem 1.3 (Escape and monodromy (source 05)). *After the substitution (51), $\Delta = c^2H$ with H irreducible and $H(0, A, B, D, E) = -108(A^2 + 4B)$. Over $\mathbb{C}$ the nonproperness set is exactly $V(cH)$, and the restriction of $\mathcal{F}$ over its complement is a finite étale cover of degree six with monodromy group S_6 . At each point of M , the hypersurface $V(H)$ has three smooth, mutually transverse analytic branches, and the local sheet exchanges form $(\mathbb{Z}/2\mathbb{Z})^3$.*

Where they are proved: part (i) in Section 4.3 and Proposition 2.1; part (ii) in Corollary 3.4 (with a second route in Section 3.4); part (iii) in Theorem 4.7; part (iv) in Corollary 4.10; the escape statements in Section 6, including Corollary 6.7; and monodromy in Theorem 8.1.

2 The polynomial map, without its large expansion

2.1 A compact definition

We reproduce the construction needed to identify the map unambiguously [3, §4.5.1]. Set

$$a_1 = xy, \quad a_2 = x^2z_1, \quad a_3 = x^3z_2, \quad a_4 = x^4z_3,$$

and apply the polynomial stage

$$\begin{aligned} \gamma &= 1 - 29a_1 + 999a_1^2 + 355a_1a_2 - 41553a_1^3 + a_4, \\ u_1 &= 1 + 27a_1 - 5a_2, \\ u_2 &= a_1 - 12a_2 + \frac{2128}{5}a_1^2 + a_3, \\ u_3 &= -4a_1 - 10a_2. \end{aligned} \tag{9}$$

Introduce sweep variables

$$w_1 = \gamma u_1, \quad w_2 = \gamma^3 u_2, \quad w_3 = \gamma^4 u_3.$$

Define potentials and one scalar polynomial by

$$\begin{aligned} G_2 &= w_2 + w_1w_3 + w_1^2 - w_1^3, \\ G_3 &= 2w_1G_2 + w_2, \\ G_4 &= w_3 - G_2^2 + 2w_1^4 - 2w_1^5, \\ X_1 &= -w_3 + 2w_2 - 2w_1 + 2w_1w_3 + 5w_1^2 - 2w_1^3 + 8w_1^4 - 10w_1^5. \end{aligned} \tag{10}$$

Put

$$\begin{aligned} S_1 &= X_1 + \gamma, & S_2 &= G_2 + w_1S_1, \\ S_3 &= G_3 + w_1^2S_1, & S_4 &= G_4 + w_2S_1. \end{aligned}$$

With $c = x\gamma$, the map is

$$\mathcal{F}(x, y, z_1, z_2, z_3) = \left(c, \frac{S_1}{c}, \frac{S_2}{c^2}, \frac{S_3}{c^3}, \frac{S_4}{c^4} \right). \tag{11}$$

All quotients in (11) cancel *as polynomials*. This cancellation is part of the construction, not a restriction to $c \neq 0$.

Proposition 2.1 (Prior polynomiality, with an independent certificate). *The expression (11) defines a polynomial map over $\mathbb{Q}$, of coordinate degrees (7, 38, 40, 42, 44), and $\det J_{\mathcal{F}} = -290$. The expanded coordinates contain respectively 6, 342, 421, 507, 904 nonzero monomials.*

Proof. Polynomiality and the degree profile are established by the stage and weighted-divisibility construction in [3, Theorem 4.5]. For an independent finite check, the accompanying script first substitutes $(w_1, w_2, w_3) = (\gamma u_1, \gamma^3 u_2, \gamma^4 u_3)$ into each S_i and divides each monomial by γ^i . It then substitutes (9), verifies that each resulting monomial has x -exponent at least i , and divides by x^i . Thus no rational evaluation or numerical cancellation is used. Counting the resulting sparse terms gives the stated data.

For the determinant, the monomial change to the a_i has determinant x^{10} . The stage has determinant -290 ; its inverse is displayed below. The change to the w_i has determinant γ^8 . Direct four-variable differentiation gives

$$\det \frac{\partial(S_1, S_2, S_3, S_4)}{\partial(\gamma, w_1, w_2, w_3)} = \gamma.$$

Padding the sweep with $c = x\gamma$ contributes another γ . Finally the target rescaling has determinant c^{-10} . On $x\gamma \neq 0$ their product is

$$x^{10}(-290)\gamma^8\gamma^2c^{-10} = -290.$$

Both sides are polynomials, so the identity holds everywhere. The small stage and sweep determinants are also independently checked in the certificate. $\square$

Second proof, by finite jets (source 05). Source 05 states this as a proposition: after the stage substitution, x^i divides $\hat{S}_i$ for $i = 1, 2, 3, 4$, and the resulting polynomial map

$$\mathcal{F} = \left(x\gamma, \frac{\hat{S}_1}{x}, \frac{\hat{S}_2}{x^2}, \frac{\hat{S}_3}{x^3}, \frac{\hat{S}_4}{x^4}\right)$$

has $\det J_{\mathcal{F}} = -290$. (This is (11), since $S_i = \gamma^i \hat{S}_i$ and $c = x\gamma$.) Here $\hat{S}_i \in \mathbb{Q}[\gamma, u_1, u_2, u_3]$ is the quotient of S_i by γ^i after the substitution $(w_1, w_2, w_3) = (\gamma u_1, \gamma^3 u_2, \gamma^4 u_3)$; the four quotients are short, and are printed in Appendix C. Two useful identities, obtained by differentiating the displayed polynomials, are

$$X_1 = -\det \frac{\partial(G_2, G_3, G_4)}{\partial(w_1, w_2, w_3)}, \quad \det \frac{\partial(S_1, S_2, S_3, S_4)}{\partial(\gamma, w_1, w_2, w_3)} = \gamma. \quad (12)$$

For divisibility by powers of x , substitute

$$\begin{aligned} \gamma &= 1 - 29xy + 999x^2y^2 + x^3(355yz_1 - 41553y^3) + x^4z_3, \\ u_1 &= 1 + 27xy - 5x^2z_1, \\ u_2 &= xy + x^2\left(-12z_1 + \frac{2128}{5}y^2\right) + x^3z_2, \\ u_3 &= -4xy - 10x^2z_1 \end{aligned}$$

into the appendix polynomials, and collect only the coefficients through x^4 . This gives

$$\hat{S}_i = x^i \beta_i + O(x^{i+1}), \quad (13)$$

where $(\beta_1, \beta_2, \beta_3, \beta_4)$ are exactly the right-hand sides of (15) for A, B, D, E . (Source 05 prints the coefficient of y^2z_1 in β_4 as $-463585/25$, which is source 02's $-92717/5$.) Here $O(x^m)$ means membership in the polynomial ideal (x^m) , not an analytic asymptotic assertion. All coefficients below x^i are zero, so this is a finite coefficient proof of divisibility; it can be performed in

$\mathbb{Q}[y, z_1, z_2, z_3][x]/(x^5)$, where terms of degree at least five cannot affect any of these claims. That is why a truncated multiplication is exact rather than heuristic.

On the nonempty open set $x\gamma \neq 0$, the map is a composition of five maps with Jacobian factors, in order,

$$x^{10}, \quad -290, \quad \gamma^8, \quad \gamma^2, \quad (x\gamma)^{-10}.$$

The first comes from the monomials a_i , the second from the stage, the third from the scaling of the sweep variables. For the fourth, the padded sweep sends $(x, \gamma, w_1, w_2, w_3)$ to $(x\gamma, S)$; expansion in the x column gives $\gamma \det JS = \gamma^2$. The last map divides S_i by $(x\gamma)^i$, keeping $x\gamma$ fixed. The product is -290 ; both sides are polynomials, and equality on a dense open set proves equality everywhere. $\square$

Remark 2.2 (Every fiber point is reduced (source 05)). In particular $\mathcal{F}$ is étale: its differential is invertible everywhere. Once finiteness of its fibers is established, every fiber point has multiplicity one. More concretely, at a fiber point the linear terms of the five defining equations generate the cotangent space, so the zero-dimensional local fiber ring is the residue field.

2.2 The stage inverse and the source hyperplane

The stage inverse is particularly useful. Given (γ, u_1, u_2, u_3) , recover

$$\begin{aligned} a_1 &= \frac{2u_1 - 2 - u_3}{58}, & a_2 &= \frac{27a_1 - u_1 + 1}{5}, \\ a_3 &= u_2 - a_1 + 12a_2 - \frac{2128}{5}a_1^2, & a_4 &= \gamma - 1 + 29a_1 - 999a_1^2 - 355a_1a_2 + 41553a_1^3. \end{aligned} \quad (14)$$

Substitution proves both compositions are the identity. Source 05 writes the second entry as $a_2 = -(4(u_1 - 1) + 27u_3)/290$, which is the same function: the first two equations solve a 2×2 linear system of determinant -290 , and the last two are triangular substitutions. Direct differentiation, in the orders (a_1, a_2, a_3, a_4) and (γ, u_1, u_2, u_3) , gives stage determinant -290 (source 05).

The polynomial restriction of $\mathcal{F}$ to $x = 0$ is

$$\begin{aligned} c &= 0, \\ A &= y, \\ B &= -\frac{4636}{5}y^2 + 29z_1, \\ D &= \frac{6039}{5}y^3 - 1041yz_1 + 5z_2, \\ E &= \frac{74000249}{25}y^4 - \frac{92717}{5}y^2z_1 + 5yz_2 - 1041z_1^2 - 2z_3. \end{aligned} \quad (15)$$

This identity can be obtained by taking the coefficients of x^i before the final divisions; the accompanying polynomial construction checks all five coordinates independently. The diagonal coefficients $1, 29, 5, -2$ are nonzero, so (15) is a triangular polynomial isomorphism. It provides exactly one source point on $x = 0$ over every target on $c = 0$. Its determinant -290 also checks the global determinant normalization.

3 The derivative identity and the uniform inverse chart

3.1 Eliminating the sweep variables

For a target b set

$$Y_1 = cA, \quad Y_2 = c^2B, \quad Y_3 = c^3D, \quad Y_4 = c^4E.$$

Write $w = w_1$. The equations $S_i = Y_i$ and (10) give

$$w_2 = Y_3 - 2wY_2 + w^2Y_1 =: V_Y(w), \quad (16)$$

$$w_3 = Y_4 + Y_2^2 - Y_1Y_3 + 2w^5 - 2w^4 =: W_Y(w). \quad (17)$$

Indeed $G_2 = Y_2 - wY_1$, and eliminating G_3 first gives (16); eliminating G_4 next gives (17). Substitution into $G_2 = w_2 + ww_3 + w^2 - w^3$ leaves

$$\begin{aligned} R_Y(w) &= 2w^6 - 2w^5 - w^3 + (Y_1 + 1)w^2 \\ &\quad + (Y_4 + Y_2^2 - Y_1Y_3 - 2Y_2 + Y_1)w + Y_3 - Y_2 = 0. \end{aligned} \quad (18)$$

This is Gao's sextic. Substituting the scaled target gives exactly P_b .

Lemma 3.1 (Derivative identity). *As a polynomial identity in Y and w ,*

$$Y_1 - X_1(w, V_Y(w), W_Y(w)) = R'_Y(w) - 2R_Y(w). \quad (19)$$

Consequently, on the fiber equation, $\gamma = R'_Y(w)$.

Proof. Insert (16) and (17) into the displayed expression for X_1 in (10) and collect coefficients in w . The result is $R'_Y - 2R_Y$. This is an identity before imposing $R_Y = 0$, so it also specifies the exact ideal membership, not just agreement at selected roots. Finally $S_1 = Y_1$ says $\gamma = Y_1 - X_1$. $\square$

Second proof, with the expansion displayed (source 05). Source 05 proves the identical lemma by printing the collected polynomial. Let $B_0 = Y_4 + Y_2^2 - Y_1Y_3$, so the linear coefficient of R_Y is $B_0 - 2Y_2 + Y_1$. Substitution into X_1 gives

$$\begin{aligned} X_1(w, V_Y(w), W_Y(w)) &= 4w^6 - 16w^5 + 10w^4 - 2w^3 + (5 + 2Y_1)w^2 \\ &\quad + (2B_0 - 4Y_2 - 2)w - B_0 + 2Y_3. \end{aligned}$$

Subtract this from Y_1 and compare with $R'_Y = 12w^5 - 10w^4 - 3w^2 + 2(Y_1 + 1)w + B_0 - 2Y_2 + Y_1$. The difference is exactly $-2R_Y$, coefficient by coefficient. In the elimination no degree drops occur: the leading coefficient of R_Y is the constant 2. $\square$

Source 05 calls this the main simplification: an arbitrary repeated root is deleted by the same condition, not merely by an extra filter checked at selected targets.

For $c \neq 0$, the relation $c = x\gamma$ forces $\gamma \neq 0$. Thus exactly the *simple* sextic roots survive. A multiple root is not a multiple point of the original fiber: it produces no source point. The distinction matters because $\mathcal{F}$ itself is everywhere étale.

3.2 A rescaling that retains the exceptional fibers

On $x \neq 0$ use the regular source function

$$v = \frac{u_1}{x} = \frac{1}{x} + 27y - 5xz_1. \quad (20)$$

When $c \neq 0$ this is w/c . The identities $P_b(cv) = c^2q_b(v)$ and $P'_b(cv) = cq'_b(v)$ suggest that the derivative of q_b , rather than a quotient by c , is the correct denominator for inversion.

Theorem 3.2 (Uniform simple-root chart). *Let*

$$\mathcal{U} = \operatorname{Spec} k[c, A, B, D, E, v, 1/q'_b(v)]/(q_b(v)).$$

There is an isomorphism over target space

$$\mathbb{A}_{x \neq 0}^5 \simeq \mathcal{U}.$$

In one direction use (20). Conversely, for a simple root v set $\delta = q'_b(v)$ and define

$$\begin{aligned} x &= \delta^{-1}, & \gamma &= c\delta, \\ u_1 &= v/\delta, & u_2 &= (D - 2vB + v^2A)/\delta^3, \\ u_3 &= (\kappa + 2cv^5 - 2v^4)/\delta^4. \end{aligned} \tag{21}$$

Recover a_i using (14), and then put

$$y = a_1\delta, \quad z_1 = a_2\delta^2, \quad z_2 = a_3\delta^3, \quad z_3 = a_4\delta^4. \tag{22}$$

No division by c occurs.

Proof. First work on $c \neq 0$. Lemma 3.1 gives $\gamma = P'_b(cv) = c\delta$, hence $x = 1/\delta$. Equations (16)–(17) give

$$w_2 = c^3(D - 2vB + v^2A), \quad w_3 = c^4(\kappa + 2cv^5 - 2v^4).$$

Dividing by γ^3 and γ^4 , respectively, gives (21). The stage inverse and monomial inverse now give (22). These steps reverse the elimination, so they prove both compositions on $c \neq 0$.

We justify extension across $c = 0$, rather than assuming it. The polynomial q_b is irreducible in $k[c, A, B, D, E, v]$: as a polynomial in E it is linear with coefficient c^3v , and that coefficient is coprime to its constant term. To see the coprimality, specialize $c = 0$ and $v = 0$:

$$q_b|_{c=0} = v^2 + Av - B, \quad q_b|_{v=0} = cD - B.$$

Neither c nor v divides the constant term. Gauss's lemma applied to the fraction field of the other variables proves irreducibility. Thus $\mathcal{U}$ is integral and its $c \neq 0$ locus is dense.

On the source open set $x \neq 0$, the functions in (20) and $\mathcal{F}$ are regular. The relations $q_b(v) = 0$ and $q'_b(v) = 1/x$ hold on the dense open set $c \neq 0$, so they hold throughout. Consequently the forward map lands in $\mathcal{U}$. All inverse formulas are regular on $\mathcal{U}$, and the two composition identities extend from its dense $c \neq 0$ locus. This proves the isomorphism everywhere. $\square$

Theorem 3.3 (Uniform reconstruction over $c \neq 0$ (source 05)). *For $c \neq 0$, the fiber over b is naturally isomorphic to*

$$\operatorname{Spec} k[w, 1/P'_b(w)]/(P_b(w)). \tag{23}$$

The bijection on points sends a simple root w to the source point

$$\begin{aligned} \gamma &= P'_b(w), & x &= c/\gamma, \\ u_1 &= w/\gamma, & u_2 &= V_Y(w)/\gamma^3, & u_3 &= W_Y(w)/\gamma^4; \end{aligned} \tag{24}$$

then (14) gives $a_1, \dots, a_4$, and $y = a_1/x$, $z_1 = a_2/x^2$, $z_2 = a_3/x^3$, $z_3 = a_4/x^4$. These formulas work in families over the entire open set $c \neq 0$.

Proof (source 05). A source point over such a target has $x\gamma \neq 0$. Its sweep parameter $w = \gamma u_1$ satisfies $P_b(w) = 0$ by the elimination, and $P'_b(w) = \gamma \neq 0$ by Lemma 3.1. Conversely, a simple root makes every denominator in (24) nonzero. The equation $P_b = 0$ recovers $G_2 = Y_2 - wY_1$; the definitions of V_Y, W_Y recover the other two potential equations; identity (19) recovers the

first sweep equation. The inverse stage and monomial substitutions give a point of the required fiber. Each operation inverts the corresponding forward operation, so this is a bijection. The same identities hold in coordinate rings after inverting c and P'_b , so they prove the scheme isomorphism (23), not merely a set-theoretic statement for closed points. For a root of multiplicity at least two, P'_b belongs to the maximal ideal of its local Artinian factor, and localization at P'_b annihilates that entire factor; a simple-root factor survives as k . $\square$

This is the $c \neq 0$ part of Theorem 3.2, with $w = cv$ and $x = c/P'_b(cv) = 1/q'_b(v)$. Source 05 treats $c = 0$ pointwise instead (Section 3.4); source 02's chart is the stronger, uniform statement.

3.3 The three effective parameters and a finite completion over $c \neq 0$ (source 05)

On $c \neq 0$ the change from Y to (a, b, d, t) , with $t = Y_2$, is a polynomial automorphism. Its inverse is

$$\begin{aligned} Y_1 &= a - 1, & Y_2 &= t, & Y_3 &= d + t, \\ Y_4 &= b - t^2 + (a - 1)(d + t) + 2t - (a - 1). \end{aligned} \quad (25)$$

Hence on $c \neq 0$ the base is $\mathbb{G}_m \times \mathbb{A}_t^1 \times \mathbb{A}_{a,b,d}^3$, and the fiber polynomial depends only on (a, b, d) . Every parameter triple is realized by actual targets, for example with $c = 1$ and $t = 0$. (This t is the parameter of the omitted surface in (42).)

There is also a finite completion over this open base:

$$\mathcal{X}^* = \operatorname{Spec} k[c, c^{-1}, t, a, b, d, w]/(P), \quad P = 2w^6 - 2w^5 - w^3 + aw^2 + bw + d. \quad (26)$$

It is finite free of rank six over the base, with basis $1, w, \dots, w^5$. Since $\partial P / \partial d = 1$, it is smooth and isomorphic to $\mathbb{G}_m \times \mathbb{A}^4$ after eliminating d . The actual source over $c \neq 0$ is exactly the open subset $P' \neq 0$ of this completion. Thus repeated-root points do exist in the finite completion: they are precisely the points excluded from the affine source by reconstruction. Over $c \neq 0$ this is the finite model Z of Section 5, since $\mathcal{N}[1/c] = \mathcal{O}[1/c]$; source 05 asked for its extension across $c = 0$, which Theorems 5.1 and 5.2 supply (Section 11.5, direction 11).

Corollary 3.4 (All geometric fibers). *For every target,*

$$\#\mathcal{F}^{-1}(b) = \#\{v : q_b(v) = 0, q'_b(v) \neq 0\} + \mathbf{1}_{c=0}. \quad (27)$$

For $c = 0$,

$$\#\mathcal{F}^{-1}(0, A, B, D, E) = \begin{cases} 3, & A^2 + 4B \neq 0, \\ 1, & A^2 + 4B = 0. \end{cases} \quad (28)$$

Proof. Partition the source into $x \neq 0$ and $x = 0$. Apply Theorem 3.2 to the first part and (15) to the second. At $c = 0$, the polynomial is $v^2 + Av - B$, with derivative $2v + A$. A nonzero discriminant gives two simple roots. A zero discriminant gives one double root, which is deleted; it contributes no point. $\square$

3.4 The exceptional hyperplane directly on the source (source 05)

Source 05 proves (28) without the rescaled chart, by splitting the source according to $x\gamma = 0$. The open-set calculation must not be specialized blindly to $c = 0$: there the sextic becomes $w^2g(w)$, independent of the target coordinates, whereas the actual fiber size is not constant.

The unique point with $x = 0$. At $x = 0$ the stage gives $\gamma = 1$, and the jets (13) give the section (15), a triangular automorphism of the hyperplane with diagonal $1, 29, 5, -2$. Its inverse is found in the order

$$\begin{aligned} y &= A, & z_1 &= \frac{B + (4636/5)y^2}{29}, & z_2 &= \frac{D + 1041yz_1 - (6039/5)y^3}{5}, \\ z_3 &= -\frac{1}{2}\left(E - 5yz_2 + 1041z_1^2 + \frac{92717}{5}y^2z_1 - \frac{74000249}{25}y^4\right). \end{aligned}$$

In particular, the origin has one explicitly visible preimage, the origin.

The branch $\gamma = 0$, $x \neq 0$. From the quotient polynomials of Appendix C,

$$(\widehat{S}_1, \widehat{S}_2, \widehat{S}_3, \widehat{S}_4)|_{\gamma=0} = (1 - 2u_1, u_1 - u_1^2, u_1^2 + u_2, u_1^4 + (1 - 2u_1)u_2 + u_3). \quad (29)$$

The first two target equations give

$$u_1 = \frac{1 - xA}{2}, \quad x^2(A^2 + 4B) = 1, \quad (30)$$

and the last two give

$$u_2 = x^3D - u_1^2, \quad u_3 = x^4E - (1 - 2u_1)u_2 - u_1^4. \quad (31)$$

For each solution $x \neq 0$ of (30), set $\gamma = 0$ and use the stage inverse (14); then recover y, z_1, z_2, z_3 by dividing the a_i by the appropriate powers of x . This supplies a unique point and proves sufficiency of every displayed equation. If $A^2 + 4B \neq 0$, the equation $x^2(A^2 + 4B) = 1$ has exactly two distinct solutions over k ; if $A^2 + 4B = 0$, it has none. Together with the single $x = 0$ point this proves (28). The two branches do not overlap, since $\gamma = 1$ on $x = 0$. (In the chart of Theorem 3.2 these two points are the simple roots $v = u_1/x = (1 - xA)/(2x)$ of $v^2 + Av - B$.)

Every fiber is therefore finite, with the claimed cardinality, and Remark 2.2 gives reducedness. With Theorem 3.3 and Section 4.3, this is source 05's proof of parts (i) and (ii) of Theorem 1.2.

4 Every fiber cardinality and the omitted surface

4.1 A finite algebraic test at every target

For a nonzero characteristic-zero polynomial f define its simple-root factor by

$$G = \gcd(f, f'), \quad S = f/G, \quad \text{simp}(f) = \frac{S}{\gcd(S, G)}, \quad (32)$$

with gcds and final polynomials normalized to be monic. A root of multiplicity m occurs in G with multiplicity $m - 1$, and in S with multiplicity one. The second gcd removes it precisely when $m \geq 2$. Hence

$$\#\{\text{simple geometric roots of } f\} = \deg \text{simp}(f).$$

Together with (27), this is an exact complete fiber-count criterion, including singular discriminant strata. It does not require factorization over an algebraic closure, and it is not limited to generic points of a discriminant component.

Source 05 gives the same test independently (source 05): with $f = P_b/2$ monic and all gcds monic, put $d_1 = \gcd(f, f')$, $h = f/d_1$ and $s_1 = h/\gcd(h, d_1)$; then $\#\mathcal{F}^{-1}(b) = \deg s_1$ for $c \neq 0$. Together with (28) this is a terminating exact algorithm for every rational or effectively represented algebraic target. It counts points over an algebraic closure and does not assert that those points are rational. It uses no numerical root-separation tolerance and no multivariate Gröbner-basis computation.

Theorem 4.1 (Cardinality spectrum). *The geometric fiber cardinalities of $\mathcal{F}$ are exactly*

$$\{0, 1, 2, 3, 4, 6\}.$$

Proof. For $c \neq 0$, P_b has degree six. Exactly five simple roots are impossible: the remaining degree one would have to be another simple root. Thus its number of simple roots belongs to $\{0, 1, 2, 3, 4, 6\}$. For $c = 0$, (28) gives only 1 and 3. Table 2 supplies exact witnesses for every listed value. For nonzero c in the table, $c = 1$, so $P_b = q_b$ with the variable renamed. The displayed factorizations and elementary gcds give the counts; the six-point row has discriminant $-993580 \neq 0$. $\square$

Corollary 4.2 (The missing five-point fiber (source 05)). *Over $c \neq 0$, a fiber has either six points or at most four points. A repeated root removes at least two of the six roots, and cannot remove just one.*

Size	Target (c, A, B, D, E)	Exact fiber polynomial or criterion
0	$(1, -11/32, 0, 25/128, 1773/4096)$	$2(w^3 - w^2/2 - w/8 - 5/16)^2$
1	$(0, 0, 0, 0, 0)$	$q(v) = v^2$; only the section remains
2	$(1, -11/8, 0, -1/16, 171/128)$	$(w^2 + 1/4)^2(2w^2 - 2w - 1)$
3	$(1, -8, 0, -7, 79)$	$(w - 1)^3(2w^3 + 4w^2 + 6w + 7)$
4	$(1, 0, 0, 0, 0)$	$w^2(w - 1)(2w^3 - 1)$
6	$(1, 0, 0, 1, 0)$	$2w^6 - 2w^5 - w^3 + w^2 + 1$

Table 2: Exact witnesses for the entire geometric cardinality spectrum.

4.2 Classifying all sextics with no surviving root

The empty-fiber case can be solved in closed form, not merely by the gcd criterion. Divide P_b by 2. Its three leading nontrivial coefficient constraints are

$$\frac{P_b(w)}{2} = w^6 - w^5 + 0w^4 - \frac{1}{2}w^3 + \text{terms of degree at most two.} \quad (33)$$

A sextic with no simple root has multiplicity partition 6, 4 + 2, 3 + 3, or 2 + 2 + 2.

Lemma 4.3 (Unique all-multiple member). *A polynomial of the form (33) has no simple root if and only if*

$$P_b(w) = 2 \left(w^3 - \frac{1}{2}w^2 - \frac{1}{8}w - \frac{5}{16} \right)^2. \quad (34)$$

The cubic in parentheses has three distinct roots.

Proof. For multiplicity 6, write $(w - r)^6$. The w^5 coefficient forces $r = 1/6$, whereas the w^4 coefficient is then $15r^2 \neq 0$.

For multiplicity 4 + 2, write $(w - r)^4(w - s)^2$. The w^5 coefficient forces $s = 1/2 - 2r$. The next two required equations are

$$-\frac{24r^2 - 8r - 1}{4} = 0, \quad \frac{8r^3 + 4r^2 - 2r + 1}{2} = 0.$$

The remainder of the second polynomial on division by the first is $(10r + 23)/36$. Thus $r = -23/10$, but $24r^2 - 8r - 1 = 3609/25 \neq 0$, a contradiction.

For multiplicity $3 + 3$, write $(w^2 - sw + p)^3$. Matching w^5 and w^4 gives $s = 1/3$ and $p = -1/9$. Its w^3 coefficient is then $5/27$, not $-1/2$.

The remaining case is a square of a monic cubic. Write $(w^3 + aw^2 + bw + d)^2$ and compare w^5, w^4, w^3 in that order. This gives $a = -1/2$, $b = -1/8$, $d = -5/16$, proving uniqueness. Conversely its square has no simple root. The cubic discriminant is $-401/128 \neq 0$, so the three roots are distinct and all have multiplicity exactly two in P_b . $\square$

Source 05 excludes more than the all-multiple partitions: it excludes every pattern with fewer than three distinct roots, and then reproves the lemma. In its words, the missing set is controlled by an unusually rigid three-parameter slice of the sextic coefficient space: the coefficients of w^5, w^4, w^3 are fixed, so the first three power sums of the roots are fixed as well. Write $Q_*(w) = w^3 - \frac{1}{2}w^2 - \frac{1}{8}w - \frac{5}{16}$ for the cubic of (34).

Lemma 4.4 (At least three distinct roots (source 05)). *Every polynomial $2w^6 - 2w^5 - w^3 + aw^2 + bw + d$ has at least three distinct roots over an algebraically closed field of characteristic zero.*

Proof. For the monic polynomial $P/2$, Newton's identities give, with roots counted with multiplicity, the power sums

$$N_1 = 1, \quad N_2 = 1, \quad N_3 = \frac{5}{2}. \quad (35)$$

If all six roots were equal to α , the first equation would give $\alpha = 1/6$, contradicting the second.

Suppose there are exactly two roots α, β , of multiplicities $m, 6 - m$. Interchange them so that $1 \leq m \leq 3$, and set $\theta = \alpha - \beta$. The first moment gives

$$\alpha = \frac{1}{6} + \frac{6 - m}{6}\theta, \quad \beta = \frac{1}{6} - \frac{m}{6}\theta.$$

The second and third moments give, respectively,

$$\theta^2 = \frac{5}{m(6 - m)}, \quad \frac{4}{9} + \frac{m(6 - m)(3 - m)}{18}\theta^3 = \frac{5}{2}.$$

Combining them yields

$$5(3 - m)\theta = 37. \quad (36)$$

For $m = 3$ this is impossible. Squaring for $m = 1$ or $m = 2$ would require $1369m(6 - m) = 125(3 - m)^2$. The two sides are 6845 and 500 for $m = 1$, and 10952 and 125 for $m = 2$. Neither equality holds. $\square$

Proposition 4.5 (The unique polynomial with no simple root (source 05)). *The polynomial $2w^6 - 2w^5 - w^3 + aw^2 + bw + d$ has no simple root if and only if*

$$(a, b, d) = \left(\frac{21}{32}, \frac{5}{32}, \frac{25}{128}\right). \quad (37)$$

At this parameter it equals $2Q_(w)^2$, and Q_* has three distinct roots.*

Second proof of Lemma 4.3 (source 05). If all roots have multiplicity at least two, Lemma 4.4 forces exactly three distinct roots, each of multiplicity two. Write the polynomial as $2(w^3 + e_1w^2 + e_2w + e_3)^2$. Comparison of the w^5, w^4, w^3 coefficients gives

$$4e_1 = -2, \quad 2(e_1^2 + 2e_2) = 0, \quad 4(e_3 + e_1e_2) = -1,$$

so $e_1 = -1/2$, $e_2 = -1/8$ and $e_3 = -5/16$. The three remaining coefficients are exactly (37). Conversely, expansion proves the square identity, and $\text{Disc } Q_* = -401/128 \neq 0$ shows that its three roots are distinct. $\square$

4.3 All seven multiplicity patterns (source 05)

Partitions of six with at least three parts are precisely the seven patterns 1^6 , 2^4 , 3^3 , $2^2 1^2$, $4 1^2$, $3 2 1$, 2^3 of Theorem 1.2; Lemma 4.4 excludes the others. A simple root contributes one point and every other root contributes zero points (Theorem 3.3). This proves the counts of Theorem 1.2(i) once all seven strata are shown to occur.

Proposition 4.6 (Every listed pattern occurs (source 05)). *Each of the seven patterns occurs in the family $2w^6 - 2w^5 - w^3 + aw^2 + bw + d$.*

Proof. The following exact choices provide five patterns.

(a, b, d)	Polynomial or property	Pattern
$(0, 1, 1)$	discriminant $-1513772 \neq 0$	1^6
$(1, 0, 0)$	$w^2(w-1)(2w^3-1)$	$2^4 1$
$(0, 0, 0)$	$w^3(2w^3-2w^2-1)$	3^3
$(-3/8, -1/8, -1/16)$	$2(w^2+1/4)^2(w^2-w-1/2)$	$2^2 1^2$
$(21/32, 5/32, 25/128)$	$2Q_*(w)^2$	2^3

The displayed cofactors are squarefree and have no unwanted common roots, as follows either by one derivative/gcd calculation or by direct substitution.

For the remaining patterns, prescribe a triple root ρ . Solving $P(\rho) = P'(\rho) = P''(\rho) = 0$ gives

$$\begin{aligned}
 a(\rho) &= -30\rho^4 + 20\rho^3 + 3\rho, \\
 b(\rho) &= 48\rho^5 - 30\rho^4 - 3\rho^2, \\
 d(\rho) &= -20\rho^6 + 12\rho^5 + \rho^3.
 \end{aligned} \tag{38}$$

At these parameters

$$\begin{aligned}
 P(w) &= (w - \rho)^3 Q_\rho(w), \\
 Q_\rho(w) &= 2w^3 + (6\rho - 2)w^2 + (12\rho^2 - 6\rho)w + 20\rho^3 - 12\rho^2 - 1,
 \end{aligned} \tag{39}$$

and in particular

$$Q_\rho(\rho) = h_4(\rho) := 40\rho^3 - 20\rho^2 - 1. \tag{40}$$

Choose a root of h_4 . Neither 0 nor $1/3$ is a root, and $\partial_w Q_\rho(\rho) = 10\rho(3\rho - 1) \neq 0$. The root ρ therefore has multiplicity exactly four in P . The other two roots cannot coincide, since that would leave only two distinct roots. This realizes $4 1^2$.

For the pattern $3 2 1$, compute

$$\text{Disc}_w Q_\rho = -4\Lambda(\rho), \quad \Lambda = 4320\rho^6 - 4320\rho^5 + 540\rho^4 - 448\rho^3 + 384\rho^2 + 36\rho + 35. \tag{41}$$

This is a nonconstant polynomial and hence has a root over k . The polynomials Λ and h_4 are relatively prime. One way to see this without a large Bézout identity is to use the fourfold-root case just established: at a root of h_4 , Q_ρ has three distinct roots, one of them ρ , because its remaining two roots are distinct and different from ρ ; thus Λ does not vanish there. Choose a root of Λ . Then Q_ρ has a repeated root and does not vanish at ρ . It cannot be a cube of one linear factor, since P would have only two distinct roots. Its pattern is therefore $2 1$, giving the pattern $3 2 1$ for P . $\square$

Every parameter triple in this proof corresponds to targets by (25). Algebraic witnesses for the one-point stratum suffice for the assertion that the stratum is nonempty; no unproved assertion about rational points on that stratum is needed. (The rows $(1, 0, 0)$, $(-3/8, -1/8, -1/16)$ and $(21/32, 5/32, 25/128)$ are the parameters of the four-, two- and zero-point targets of Table 2. The row $(0, 0, 0)$ is a different three-point witness from that table's, whose parameter is $(-7, 15, -7)$. The row $(0, 1, 1)$ is the parameter of the second arithmetic target b_{**} of Section 8.4.)

Theorem 4.7 (Exact omitted locus). *Over an algebraically closed field, $\mathbb{A}^5 \setminus \mathcal{F}(\mathbb{A}^5)$ is exactly the surface M of (7). It is closed, smooth, irreducible, and isomorphic to $\mathbb{G}_m \times \mathbb{A}^1$. An explicit parametrization, with $c \neq 0$ and $t \in k$, is*

$$\begin{aligned} A &= -\frac{11}{32c}, & B &= \frac{t}{c^2}, \\ D &= \frac{t + 25/128}{c^3}, & E &= \frac{-t^2 + (53/32)t + 1773/4096}{c^4}. \end{aligned} \quad (42)$$

Proof. There are no empty fibers on $c = 0$ by (15). For $c \neq 0$, Lemma 4.3 applies. The coefficients of $w^2, w, 1$ in its right-hand side are $21/32, 5/32, 25/128$. Comparing these with (4) gives exactly (7).

The first equation of (7) forces c to be invertible even scheme-theoretically: $c(-32A/11) = 1$ in the coordinate ring. With $t = c^2B$, the other equations solve uniquely to (42). Thus the coordinate ring is $k[c, c^{-1}, t]$. This proves all assertions, including closedness in $\mathbb{A}^5$, dimension two, and codimension three. $\square$

Second proof, through the effective parameters (source 05). No target with $c = 0$ is omitted, by Section 3.4. For $c \neq 0$, Theorem 3.3 and Proposition 4.5 show that a target is omitted exactly when its effective parameters equal (37). Inserting these constants into (25) gives

$$Y_1 = -\frac{11}{32}, \quad Y_2 = t, \quad Y_3 = t + \frac{25}{128}, \quad Y_4 = -t^2 + \frac{53}{32}t + \frac{1773}{4096},$$

and dividing by c^i gives (42). The same set is cut out globally by three polynomial equations, which are (7) cleared of denominators:

$$\begin{aligned} 32cA + 11 &= 0, \\ 128(c^3D - c^2B) - 25 &= 0, \\ 32[c^4(E + B^2 - AD) - 2c^2B + cA] - 5 &= 0. \end{aligned} \quad (43)$$

The first equation forces c to be invertible, with $c^{-1} = -32A/11$. On that open chart these equations are precisely $a = 21/32$, $d = 25/128$ and $b = 5/32$, and the coordinate change (25) proves that the quotient ring is $k[c, c^{-1}, t]$. This proves closedness in $\mathbb{A}^5$, not merely closedness in $c \neq 0$, and proves smoothness, the dimension, and that M is a complete intersection. $\square$

Remark 4.8 (Geometric image versus rational image). The equality $\mathcal{F}(\mathbb{A}^5) = \mathbb{A}^5 \setminus M$ is a statement about geometric points. Over $\mathbb{Q}$, a surviving simple root need not be rational. There are many rational targets outside M with no rational preimage; the explicit S_6 specialization in Section 8 is one of them.

Remark 4.9 (An explicit rational omitted point (source 05)). Source 05 states as a corollary that the rational target

$$\left(1, -\frac{11}{32}, 0, \frac{25}{128}, \frac{1773}{4096}\right) \quad (44)$$

has no preimage even over $\mathbb{C}$; it is the zero-point witness of Table 2 ($c = 1, t = 0$ in (42)). This is an absolute geometric omission, not merely a failure to find rational preimages. By contrast, Sections 8 and 8.4 give rational targets with six complex preimages but no rational preimage.

Corollary 4.10 (An obstruction to low-dimensional stabilization (source 05)). *The map $\mathcal{F}$ is not polynomial-equivalent to $\Phi \times \text{id}_{\mathbb{A}^{5-m}}$ for any polynomial self-map Φ of $\mathbb{A}^m$ with $m \leq 3$, regardless of its geometric degree.*

Proof. Polynomial equivalence means composition with polynomial automorphisms on the source and on the target. A source automorphism does not change the image; a target automorphism carries its complement isomorphically to the new complement. For a product map that complement is

$$\Omega_\Phi \times \mathbb{A}^{5-m}, \quad \Omega_\Phi = \mathbb{A}^m \setminus \text{im } \Phi.$$

If such an equivalence existed, this product would be isomorphic to the nonempty irreducible surface M . The set Ω_Φ is closed: intersect the closed product complement with the section having identity coordinates zero. Dimension then gives $\dim \Omega_\Phi = 2 - (5 - m) = m - 3$. For $m < 3$ this is impossible. For $m = 3$, irreducibility forces the reduced zero-dimensional variety Ω_Φ to be one point, and the product would be $\mathbb{A}^2$. But $M \simeq \mathbb{G}_m \times \mathbb{A}^1$ is not isomorphic to $\mathbb{A}^2$: its coordinate ring has the nonconstant unit c , whereas every unit of $k[y_1, y_2]$ is constant, and an isomorphism of varieties preserves the group of units of its coordinate ring. This contradiction proves the claim and Theorem 1.2(iv). $\square$

The obstruction uses more than geometric degree: it excludes a stabilized three-dimensional map even when that map also has geometric degree six. *A possible four-dimensional stabilization is not excluded by this argument* (Section 11.5, direction 13). Source 05's $N = \mathbb{A}^m \setminus \text{im } \Phi$ is written Ω_Φ here, since $\mathcal{N} = \mathcal{N}$ is the normalization algebra and $\mathcal{O} = \mathcal{O}$ the monogenic order.

4.4 Why codimension three is natural here (source 05)

The family of fiber polynomials has three effective parameters (a, b, d) . At the omitted parameter the sextic has three independent double roots. Prescribing all three double-root degenerations uses all three transverse parameter directions; the remaining target parameters c and t do not occur in the fiber polynomial. That dimension count becomes a proof once Proposition 4.5 shows that no other all-multiple-root pattern is possible.

The result does not say that codimension three is a universal maximum for omitted loci of Keller maps; it says that this explicit map realizes it. Nor should the codimension-three omitted set be confused with the nonproperness set, which is a divisor (Section 6).

5 An explicit finite normalization

5.1 The monogenic order is not normal

Let

$$\mathcal{A} = k[c, A, B, D, E], \quad \mathcal{O} = \mathcal{A}[w]/(P_b(w)).$$

Since 2 is a unit, $\mathcal{O}$ is free of rank six over $\mathcal{A}$. The polynomial P_b is irreducible: as a polynomial in E , its leading coefficient is $c^4 w$, and its constant term is divisible by neither c nor w . Gauss's lemma gives the assertion, as in the proof of Theorem 3.2. Thus $\mathcal{O}$ is a domain.

The source function field is its fraction field. Indeed, away from $c = 0$ the root chart identifies $w = cv$ and gives the full source coordinates rationally in b, w . Nevertheless, $\mathcal{O}$ is not normal along its collapsed locus $c = w = 0$. At $c = 0$ its defining equation is

$$P_b(w) = w^2 g(w),$$

independent of A, B, D, E . That single double root has forgotten the quadratic $v^2 + Av - B$ which controls the two additional source branches.

5.2 One fraction and six basis elements

Inside $\text{Frac}(\mathcal{O})$ put

$$\zeta = \frac{wJ(w)}{c}, \quad \mathcal{N} = \mathcal{A}[w, \zeta]. \quad (45)$$

The polynomial relation $P_b = 0$ gives the symmetric identities

$$wJ = c\zeta, \quad w(\zeta - T) = cL, \quad \zeta(\zeta - T) = LJ. \quad (46)$$

They are exactly the multiplication rules (8).

Theorem 5.1 (Finite free normalization). *The algebra $\mathcal{N}$ is the integral closure of both $\mathcal{O}$ and $\mathcal{A}$ in the source function field. It is free over $\mathcal{A}$ with basis*

$$\mathcal{B} = (1, w, w^2, w^3, w^4, \zeta).$$

The affine variety $Z = \text{Spec } \mathcal{N}$ is smooth over k , and $\pi : Z \rightarrow \mathbb{A}^5$ is finite flat of degree six.

Proof. Finite freeness. The $\mathcal{A}$ -span of $\mathcal{B}$ contains 1, w and ζ . It is stable under multiplication by w and by ζ : use the three rules (8), reducing w^5 , $w\zeta$, and ζ^2 . This span is therefore the whole algebra $\mathcal{N}$. The six elements are independent, because relative to $1, w, \dots, w^5$ the last element has coefficient $2/c$ on w^5 . The degree-six irreducibility of P_b gives independence over $\text{Frac}(\mathcal{A})$, and hence over $\mathcal{A}$. Thus $\mathcal{N}$ is finite free of rank six. The same reduction and independence show that the three displayed relations form a presentation: any polynomial in w, ζ reduces to the basis span, and no additional relation can remain there. In particular its elements are integral over $\mathcal{A}$.

Three useful charts. On $c \neq 0$ we have $\mathcal{N}[1/c] = \mathcal{O}[1/c]$. On $w \neq 0$, the second relation in (46) gives $\zeta = T + cL/w$, so $\mathcal{N}[1/w] = \mathcal{O}[1/w]$. On $J \neq 0$ put $v = \zeta/J$. The first relation gives $w = cv$, and the third gives

$$v^2J(cv) - Tv - L = 0.$$

Conversely $w = cv$ and $\zeta = vJ(cv)$ satisfy all three relations. Therefore

$$\mathcal{N}[1/J] \simeq \mathcal{A}[v, 1/J(cv)]/(q_b(v)). \quad (47)$$

The three opens cover Z : a point with $c = w = 0$ has $J = 1$.

Smoothness on those charts. On $c \neq 0$, work in the hypersurface $P_b = 0$. Its derivative with respect to E is c^4w . If that derivative vanishes, then $w = 0$, and the derivative with respect to D is c^3 , which is nonzero. Thus this open is smooth. At a point with $c = 0, w \neq 0$, the equation forces $g(w) = 0$ and

$$\frac{\partial P_b}{\partial w} = w^2g'(w) \neq 0,$$

since $g = (w - 1)(2w^3 - 1)$ has four distinct nonzero roots. Finally, at a point with $c = w = 0$, chart (47) applies and $\partial q_b / \partial B = -1$. These checks prove smoothness everywhere.

A smooth integral algebra over a characteristic-zero field is normal. The finite integral algebra $\mathcal{N}$ has the same fraction field as $\mathcal{O}$. Any element of that field integral over $\mathcal{O}$ is integral over $\mathcal{N}$ and therefore belongs to $\mathcal{N}$. This proves that $\mathcal{N}$ is its integral closure. Since $\mathcal{O}$ is itself integral over $\mathcal{A}$, it is also the integral closure of $\mathcal{A}$ in that field. $\square$

5.3 The exact fiber of the finite model at $c = 0$

Reducing (46) modulo c gives

$$wg(w) = 0, \quad w(\zeta + A) = 0, \quad \zeta(\zeta + A) = Bg(w).$$

As $g(0) = 1$, the zero locus of w and the zero locus of g are disjoint. The Chinese remainder decomposition is

$$\mathcal{N}/(c) \simeq k[A, B, D, E, \zeta]/(\zeta^2 + A\zeta - B) \times k[A, B, D, E, w]/(g(w)). \quad (48)$$

On the second factor $\zeta = -A$; on the first $w = 0$. The fiber has length two from the quadratic factor and length four from the residual quartic factor. This accounts for all six sheets, even where the quadratic becomes nonreduced.

Theorem 5.2 (The source as an open part of its finite model). *The map $\mathcal{F}$ factors as an open immersion followed by the finite map:*

$$\mathbb{A}^5 \xrightarrow{j} Z \xrightarrow{\pi} \mathbb{A}^5.$$

Over an algebraically closed field, let ρ_1, ρ_2, ρ_3 be the three roots of $2w^3 = 1$, and let

$$E_i = \{c = 0, w = \rho_i, \zeta = -A\} \subset Z.$$

Each E_i is isomorphic to $\mathbb{A}^4$, and π maps it isomorphically onto $c = 0$. The exact complement of the source is

$$Z \setminus j(\mathbb{A}^5) = \text{Ram}(\pi) \cup E_1 \cup E_2 \cup E_3. \quad (49)$$

The three E_i are unramified and disjoint from $\text{Ram}(\pi)$.

Proof. On the source define $w = \gamma u_1$. The rational function ζ of (45) is regular there. On $x \neq 0$ it equals $vJ(cv)$ by the root chart. On $w \neq 0$ it equals $T + cL/w$. These opens cover the source, because $x = 0$ implies $\gamma = u_1 = w = 1$. Thus they give a morphism $j : \mathbb{A}^5 \rightarrow Z$ over the target.

For $c \neq 0$, π is the sextic root cover. Its unramified points are exactly the simple roots, and Theorem 3.2 identifies all of these with source points. For $c = 0, w = 0$, decomposition (48) and chart (47) identify the unramified points with the simple roots of $v^2 + Av - B$; all of these are again source points. The root $w = 1$ at $c = 0$ is unramified, and is precisely the image of the source hyperplane under (15). The other three quartic roots are also unramified, but cannot come from the source: $x = 0$ gives $w = 1$, whereas $x \neq 0$ and $c = 0$ give $\gamma = 0$ and $w = 0$.

This proves the asserted set-theoretic image and shows that j is injective on geometric points. It lands in the étale locus of π . Both its source and that locus are étale over the target; a morphism between such schemes is étale [4, Tag 02GH]. Geometric injectivity makes it universally injective. An étale universally injective morphism is an open immersion [5, Tag 025G]. Finally the four nonzero roots of g are simple everywhere over $c = 0$, so all three removed components are unramified and disjoint from the ramification locus. The description of each E_i follows directly from (48). $\square$

The phrase “three escapes” refers precisely to these three missing, unramified sheets over the generic point of $c = 0$. At a generic point of the other exceptional component, the mechanism is different: two sheets meet at one ramification point, and that point is removed from the source.

6 The discriminant and the nonproperness locus

6.1 Removing a spurious discriminant factor

Let

$$\Delta(a, b, d) = \text{Disc}_w(2w^6 - 2w^5 - w^3 + aw^2 + bw + d). \quad (50)$$

It is unnecessary to print its full expansion. The target substitution is

$$a = 1 + cA, \quad b = -cT = c^4\kappa - 2c^2B + cA, \quad d = -c^2L = c^3D - c^2B. \quad (51)$$

Theorem 6.1 (Normalized discriminant). *The expression*

$$H = \frac{\Delta(1 + cA, -cT, -c^2L)}{c^2} \quad (52)$$

is a polynomial, irreducible over every algebraically closed field of characteristic zero, and

$$H|_{c=0} = -108(A^2 + 4B). \quad (53)$$

In the basis $\mathcal{B}$ of Theorem 5.1, the trace pairing of $\mathcal{N}/\mathcal{A}$ has determinant $H/256$.

Proof. The discriminant of the power basis of the monic polynomial $P_b/2$ is $\text{Disc}(P_b)/2^{10}$. The change from that power basis to $\mathcal{B}$ has determinant $2/c$. A trace discriminant changes by the square of the basis determinant. Therefore

$$\text{Disc}(\mathcal{B}) = \frac{4}{c^2} \frac{\text{Disc}(P_b)}{2^{10}} = \frac{H}{256}. \quad (54)$$

The left-hand side belongs to $\mathcal{A}$ because all multiplication rules have polynomial coefficients. This proves polynomiality of H .

For its value at $c = 0$, first assume $A^2 + 4B \neq 0$. The two roots of P_b near zero have the form $cv_+ + O(c^2)$ and $cv_- + O(c^2)$, where $v_{\pm}$ are the two roots of $v^2 + Av - B$. Their squared difference contributes $c^2(A^2 + 4B)$ to first order. The other four roots tend to the distinct roots of g . The leading-coefficient factors and all cross-differences combine to $\text{Disc}(g) = -108$: explicitly, the product of the four roots of g is $1/2$, the cross-difference contribution is $(1/2)^4$, and the leading-coefficient ratio is $2^{10}/2^6 = 16$. Thus the coefficient of c^2 in $\text{Disc}(P_b)$ is $-108(A^2 + 4B)$. This formal expansion is valid on a dense open set and therefore gives the polynomial identity (53). It also follows by direct substitution into the exact resultant.

For irreducibility, the double-root incidence of the polynomial in (50) is parametrized by (r, a) :

$$\begin{aligned} b &= -12r^5 + 10r^4 + 3r^2 - 2ar, \\ d &= 10r^6 - 8r^5 - 2r^3 + ar^2. \end{aligned} \quad (55)$$

Its image is exactly the discriminant hypersurface; hence that hypersurface is irreducible. The discriminant is reduced, not a higher power: at $(a, b, d) = (1, 0, 0)$ the polynomial is $w^2(w - 1)(2w^3 - 1)$ with one ordinary double root, and varying d gives a simple zero of the discriminant. For example the coefficient of d in $\Delta(1, 0, d)$ is $432 \neq 0$.

After inverting c , substitution (51) is a coordinate isomorphism

$$(c, A, B, D, \kappa) \longleftrightarrow (c, B, a, b, d).$$

Therefore H is irreducible after inverting c . Any additional factor before localization would have to be a power of c . Equation (53) excludes such a factor. This proves irreducibility. $\square$

For a finite locally free algebra in characteristic zero, the trace pairing is nondegenerate precisely on the étale locus [6, Tag 0BJF]. Thus $V(H)$ is the branch locus of the finite normalization, while the factor c^2 in the unnormalized sextic discriminant records its nonnormal collapse. The supplied sparse file stores H with 179 monomials in (c, A, B, D, κ) .

6.2 The discriminant surface and its normalization (source 05)

Let $\mathcal{D} = V(\Delta) \subset \mathbb{A}_{a,b,d}^3$ be the discriminant surface in coefficient space. Writing it as a discriminant fixes its scalar normalization; no choice of a reduced defining equation is hidden in later constants. Source 05 ships the expanded 54-term polynomial Δ (Appendix A). The incidence (55) used for irreducibility in Theorem 6.1 is in fact the normalization of $\mathcal{D}$, with r written ρ :

Proposition 6.2 (An explicit normalization of $\mathcal{D}$ (source 05)). *The polynomial Δ is absolutely irreducible up to a nonzero scalar. The normalization of $\mathcal{D}$ is $\mathbb{A}_{a,\rho}^2$, with finite normalization map*

$$(a, \rho) \mapsto (a, -12\rho^5 + 10\rho^4 + 3\rho^2 - 2a\rho, 10\rho^6 - 8\rho^5 - 2\rho^3 + a\rho^2). \quad (56)$$

The points above a parameter are exactly its distinct repeated roots.

Proof. Solving $P(\rho) = P'(\rho) = 0$ for b, d gives (56). Thus the repeated-root incidence variety is the affine plane with coordinates (a, ρ) . Its projection is finite: ρ is integral over the coefficient ring, since it satisfies the monic equation $P(\rho)/2 = 0$. The image is exactly the discriminant zero set, so that set is irreducible.

At $(a, b, d) = (1, 0, 0)$ the polynomial has exactly one double root and four simple roots, and direct differentiation of the discriminant gives $\partial_d \Delta(1, 0, 0) = 432 \neq 0$. Thus Δ is reduced along its unique irreducible component: a polynomial with irreducible zero set is a scalar times a power of the irreducible defining polynomial, and the nonzero derivative forces that power to be one.

The condition $\deg \gcd(P, P') = 1$ is open in the discriminant, by the rank conditions on the Sylvester matrix. It holds at the displayed witness and is therefore dense. On this locus the projection has one-point fibers, so in characteristic zero it is birational. A finite birational map from the normal affine plane is the normalization. The description of its fibers follows from its incidence definition. $\square$

For example, the normalization has three distinct points above the omitted parameter (37). A single irreducible discriminant surface can therefore have three local analytic branches at that parameter: global irreducibility and local reducibility are different notions.

Second proof of (53) and of the irreducibility of H , by coprime factor lifting (source 05). At $c = 0$ the sextic is $w^2 g(w)$, with $g(0) = 1$ and $\text{Disc } g = -108$. Work in the formal series ring $k[A, B, D, E][[c]]$. Coprime factor lifting splits P_b uniquely into a monic quadratic factor ϕ_c reducing to w^2 and a degree-four factor ψ_c reducing to g ; only their initial terms are needed. The effective parameters have expansions

$$a = 1 + cA, \quad b = cA - 2c^2B + O(c^4), \quad d = -c^2B + c^3D.$$

The quadratic factor has linear coefficient $cA + O(c^2)$ and constant coefficient $-c^2B + O(c^3)$, so $\text{Disc } \phi_c = c^2(A^2 + 4B) + O(c^3)$. The product formula for discriminants gives $\text{Disc } P_b = (\text{Disc } \phi_c)(\text{Disc } \psi_c) \text{Res}(\phi_c, \psi_c)^2$, and at $c = 0$ the last two factors are -108 and 1 . Consequently

$$\Delta(a, b, d) = -108c^2(A^2 + 4B) + O(c^3). \quad (57)$$

As an identity about the first three coefficients of a polynomial in c , this proves divisibility by c^2 and shows that c^3 does not divide it; the supplied script checks the same coefficient identity directly. After inverting c , the coefficient change identifies the ring with $k[c, c^{-1}, t, a, b, d]$, and Proposition 6.2 shows that the substituted discriminant is irreducible there. Any factorization of H in the polynomial ring would therefore have a factor that becomes a unit after inverting c , hence a scalar times a power of c , which (57) rules out. $\square$

6.3 Two components of nonproperness

Definition 6.3. For a polynomial map $f : \mathbb{C}^n \rightarrow \mathbb{C}^n$, its nonproperness set consists of targets b for which no neighborhood V of b has $f^{-1}(V) \rightarrow V$ proper. Equivalently, there exists an unbounded sequence of source points whose images converge to b .

Theorem 6.4 (Exact nonproperness locus). *For $\mathcal{F} : \mathbb{C}^5 \rightarrow \mathbb{C}^5$,*

$$\text{NP}(\mathcal{F}) = V(cH) = V(c) \cup V(H). \quad (58)$$

These are two distinct irreducible hypersurfaces. Their intersection is

$$c = 0, \quad A^2 + 4B = 0. \quad (59)$$

The generic fiber on $V(c)$ has three points; the generic fiber on $V(H)$ has four points; every point of their intersection has one point in its original fiber. (Source 05:) Both components are smooth along (59) and meet transversely there; smoothness of $V(H)$ is asserted along that intersection, not everywhere.

Proof. If $cH \neq 0$, the sextic has six distinct roots and the original map has six points over the target. Since $\mathcal{F}$ has nonzero Jacobian, choose six disjoint local inverse neighborhoods. On a sufficiently small common target neighborhood they supply six inverse branches. There can be no additional branch, because every fiber has at most six points. After shrinking to a closed ball inside this neighborhood, the preimage is a finite union of compact continuous images of that ball. Thus $\mathcal{F}$ is proper near the target.

At a target on $V(cH)$, the fiber has fewer than six points by Corollary 3.4. Suppose the map were proper over a small connected neighborhood. A proper local homeomorphism with finite fibers is a covering onto a union of components; its fiber cardinality is locally constant. Nearby points with $cH \neq 0$ have six preimages, a contradiction. This proves (58).

Irreducibility and the intersection statement follow from Theorem 6.1. At a generic point of the discriminant there is one double root and four simple roots, so the original fiber has four points. The hyperplane and intersection counts are (28). For transversality (source 05): by (53), $\partial H / \partial B = -432$ on $c = 0$, which is nonzero at every point of (59). The gradients of c and H are therefore linearly independent there, proving smoothness and transversality. $\square$

Second proof of (58), through the finite completion (source 05). On $U = \{c \neq 0, \Delta \neq 0\}$ the sextic is squarefree. In the rank-six algebra of (26), P' is invertible, because a Bézout identity between P and P' exists after inverting their resultant. The entire finite completion is therefore the actual source over U , so $\mathcal{F}^{-1}(U) \rightarrow U$ is finite étale of degree six, and proper. At every point outside U the fiber-size classification gives fewer than six points: one or three on $c = 0$, and at most four on $c \neq 0, \Delta = 0$. Suppose the map were proper over a neighborhood of such a point. An everywhere local biholomorphism that is proper over a sufficiently small connected neighborhood is a finite covering there; its number of sheets equals the size of its central fiber and is locally constant. The neighborhood also contains points of U , whose fibers have six points, a contradiction. This argument includes omitted points: a proper covering with zero sheets would have empty image on the neighborhood. $\square$

The four nearby fiber counts at a point of the intersection (59) are worth distinguishing (source 05):

location	fiber size
$c \neq 0, H \neq 0$	6
$c = 0, H \neq 0$	3
$c \neq 0, H = 0$ near the intersection	4
$c = 0, H = 0$	1

The third row holds because $V(H)$ is smooth near the intersection, so the off-hyperplane sextic has exactly one double root there. In this local picture three sheets are lost on one divisor and two on the other; the one visible hyperplane point remains at their intersection.

There is an equivalent finite-model explanation. The image under π of the ramification locus is $V(H)$, whereas the three deleted unramified components all map to $V(c)$. Equation (49) therefore describes exactly which points of the finite cover have to be added to restore properness. The omitted surface M is much smaller than the nonproperness set: points of M lose every sheet, while a generic nonproper target retains several sheets.

6.4 Every local analytic type of the discriminant (source 05)

The three-parameter slice (a, b, d) is not accidentally too small to see the relevant degenerations: it is transverse to every contact pattern that actually occurs. This gives a local analytic description at every point of $\mathcal{D}$, not only at the omitted parameter. (This partly answers source 02's question 3 in Section 11.)

Lemma 6.5 (Hermite transversality (source 05)). *Suppose $P = 2w^6 - 2w^5 - w^3 + a_0w^2 + b_0w + d_0$ has distinct repeated roots $\rho_1, \dots, \rho_j$ of multiplicities $m_1, \dots, m_j$, and put $\ell = \sum_{i=1}^j (m_i - 1)$. Then $\ell \leq 3$, and the perturbations $\delta a w^2 + \delta b w + \delta d$ map surjectively to*

$$\bigoplus_{i=1}^j \mathbb{C}[w]/(w - \rho_i)^{m_i-1}. \quad (60)$$

Proof. The sum of multiplicity excesses is six minus the number of distinct roots, so Lemma 4.4 gives $\ell \leq 3$. The Chinese remainder theorem identifies (60) with $\mathbb{C}[w]/\prod_i (w - \rho_i)^{m_i-1}$. The modulus has degree $\ell \leq 3$, so every residue class has a representative of degree at most two. This is precisely the asserted surjectivity. $\square$

Theorem 6.6 (Local analytic normal forms (source 05)). *In suitable local analytic coordinates $(\sigma_1, \sigma_2, \sigma_3)$ on the coefficient space $\mathbb{A}_{a,b,d}^3$, Δ is a nonvanishing analytic unit times the equation in the following table; unused coordinates are smooth parameters.*

Multiplicity pattern	Local equation for $\mathcal{D}$	Fiber size
1^6	no discriminant point	6
$2\,1^4$	σ_1	4
$3\,1^3$	$4\sigma_1^3 + 27\sigma_2^2$	3
$2^2\,1^2$	$\sigma_1\sigma_2$	2
$4\,1^2$	$\Delta_4(\sigma_1, \sigma_2, \sigma_3)$	2
$3\,2\,1$	$\sigma_3(4\sigma_1^3 + 27\sigma_2^2)$	1
2^3	$\sigma_1\sigma_2\sigma_3$	0

Here Δ_4 is the discriminant of $\omega^4 + \sigma_1\omega^2 + \sigma_2\omega + \sigma_3$,

$$\Delta_4 = 256\sigma_3^3 - 128\sigma_1^2\sigma_3^2 + 144\sigma_1\sigma_2^2\sigma_3 - 27\sigma_2^4 + 16\sigma_1^4\sigma_3 - 4\sigma_1^3\sigma_2^2. \quad (61)$$

In particular, the singular points of $\mathcal{D}$ are exactly the parameters with a root of multiplicity at least three or with at least two distinct repeated roots.

Proof. Choose disjoint small disks about the distinct roots of the reference polynomial. In a neighborhood of its coefficients, analytic factorization groups the roots in each disk into a monic factor of the corresponding degree. This follows, for example, from the implicit function theorem for multiplication of coprime monic factors, whose linearization is invertible by the polynomial Chinese remainder theorem. The resultants between different factors do not vanish in this neighborhood,

so the discriminant product formula expresses Δ , up to an analytic unit, as the product of the discriminants of the repeated-root factors; simple-root factors contribute only units.

Center the degree- m_i factor about its mean root to remove its degree- $(m_i - 1)$ term. Its remaining deformation coefficients have tangent space $\mathbb{C}[\omega]/\omega^{m_i-1}$. The differential from the original perturbation $h(w)$ to these coefficients is the class of h at ρ_i , multiplied by the inverse of the other factors; this multiplier is a unit in the corresponding truncated local ring. The combined differential is therefore surjective by Lemma 6.5. Complete these ℓ deformation coefficients to three analytic coordinates by the inverse function theorem.

Thus the repeated-root factors, after centering, are independent universal monic polynomials of degrees m_i with their degree- $(m_i - 1)$ coefficients removed. A quadratic gives one coordinate as its discriminant; a centered cubic gives $-4\sigma_1^3 - 27\sigma_2^2$; a centered quartic gives (61). Multiplying these factors yields the table, with harmless nonzero constants absorbed into units. In the smooth double-root case the differential is nonzero; all other displayed equations have zero differential at their base point. This proves the singular-locus criterion as well. $\square$

6.5 Three transverse escape branches along the omitted surface (source 05)

Corollary 6.7 (Normal crossings and local monodromy (source 05)). *Near any point of M there are analytic coordinates $(\epsilon_1, \epsilon_2, \epsilon_3, \lambda_1, \lambda_2)$ on the target for which*

$$V(H) = V(\epsilon_1\epsilon_2\epsilon_3), \quad M = V(\epsilon_1, \epsilon_2, \epsilon_3).$$

If exactly j of the ϵ_i vanish, the fiber has $6 - 2j$ points. A small loop around $\epsilon_i = 0$ exchanges one pair of sheets and fixes the other four. The local monodromy image is $(\mathbb{Z}/2\mathbb{Z})^3$, acting on three disjoint pairs.

Proof. At the omitted parameter, $P = 2Q_\star^2$ and $Q_\star$ has three distinct roots. The 2^3 row of Theorem 6.6 gives the product equation; the two extra coordinates come from c and t (Section 3.3, $c \neq 0$ on M). More explicitly, each of the three quadratic root clusters can be centered and written as $\omega_i^2 - \epsilon_i$. Its two roots are distinct and survive reconstruction when $\epsilon_i \neq 0$; when $\epsilon_i = 0$, its double root is entirely removed by localization at P' . The three clusters stay disjoint, so exactly two points are lost for each zero coordinate. Analytic continuation of $\sqrt{\epsilon_i}$ around zero exchanges the two roots in that cluster and no others. These three transpositions commute and are independent. $\square$

This also explains why the empty fiber is not produced by six source points colliding at one affine source point. The differential of $\mathcal{F}$ is invertible everywhere. The collisions occur in the finite completion (26) (equivalently in Z), and the collision points have been removed from the affine source: with c tending to a nonzero value, already the coordinate $x = c/P'(w)$ diverges as $P'(w)$ tends to zero. (This matches source 02's question 4, which asks for the escape rates.)

Together, Theorems 6.4 and 6.6 give all local analytic types of the nonproperness divisor. Off $c = 0$ they are the types in the table, with two smooth factors. On $c = 0$ the divisor is smooth away from (59), and has two transverse branches along that intersection.

7 A general quadratic-contraction normalization lemma

The short normalization proof is not peculiar to the large coefficients in the stage. Its algebraic core applies to a wider class of degenerating root covers. The following lemma records that mechanism without claiming that every such cover is a Keller map.

Theorem 7.1 (Quadratic-contraction algebra). *Let A_0 be a domain, let $c \in A_0$ be nonzero, and let $J(w) \in A_0[w]$ have degree $m \geq 1$ and leading coefficient $a_m \in A_0^\times$. Choose $T, L \in A_0$ and suppose*

$$P(w) = w^2 J(w) - cTw - c^2 L$$

is irreducible over $\text{Frac}(A_0)$. In its root field define $\zeta = wJ(w)/c$ and $N = A_0[w, \zeta]$. Then:

- (i) *N is finite free of rank $m + 2$ over A_0 , with basis $1, w, \dots, w^m, \zeta$ and relations*

$$wJ = c\zeta, \quad w\zeta = Tw + cL, \quad \zeta^2 = T\zeta + LJ.$$

- (ii) *Its trace discriminant in that basis equals*

$$\frac{\text{Disc}(P)}{a_m^{2m} c^2}.$$

- (iii) *On $c \neq 0$ or $w \neq 0$ the algebra equals the original monogenic order. On $J \neq 0$ it has the chart*

$$A_0[v, 1/J(cv)]/(v^2 J(cv) - Tv - L).$$

If c and $J(0)$ generate the unit ideal, these three opens cover $\text{Spec } N$. If their rings are normal, N is the integral closure of the monogenic order in its fraction field.

Proof. The three relations follow by multiplying $P = 0$ by suitable powers of J/c and cancelling in the fraction field. They show that the span of the displayed basis is closed under multiplication by w and ζ ; the leading coefficient a_m is a unit, so w^{m+1} reduces using $wJ = c\zeta$. Independence follows from the coefficient a_m/c of w^{m+1} in ζ . This proves (i).

The monic power-basis discriminant is $\text{Disc}(P)/a_m^{2m+2}$. The basis determinant is a_m/c , giving (ii). The chart calculations are the same elementary substitutions as in (47). If $c = w = J = 0$, then $c = J(0) = 0$, which is excluded by the unit-ideal assumption. Thus the charts cover. Normality is local; a finite normal algebra in the same fraction field is the integral closure. This proves (iii). $\square$

For F_6 , take $m = 4$, $a_m = 2$, and $J(0) = 1 + cA$. The smoothness checks in Theorem 5.1 supply the normality hypotheses. For a general J, T, L , those hypotheses must be checked: the finite free algebra in this lemma is not automatically normal. This distinction is important when attempting degree-twelve or higher analogues.

The lemma explains the otherwise mysterious square factor c^2 in a root discriminant: one basis vector must be divided by c to separate the two collapsed directions. It also provides a practical alternative to normalizing a large graph ideal in all source and target variables.

8 Geometric monodromy and an arithmetic radical obstruction

8.1 A Morse slice with full symmetric monodromy

Set

$$p(w) = 2w^6 - 2w^5 - w^3 + w^2.$$

On the target line $(c, A, B, D, E) = (1, 0, 0, t, 0)$, the fiber polynomial is $p(w) + t$. Its discriminant is

$$\text{Disc}_w(p(w) + t) = -4t(32t + 3)(11664t^3 - 5680t^2 + 1149t - 36). \quad (62)$$

The polynomial on the right is squarefree. This can be checked by the Euclidean algorithm, and is independently verified in the code.

Theorem 8.1 (Geometric S_6). *The six-sheeted cover over $\mathbb{C}^5 \setminus V(cH)$ has monodromy group S_6 . Equivalently, P_b has Galois group S_6 over $\mathbb{C}(c, A, B, D, E)$.*

Proof. For a polynomial p of degree six, the discriminant of $p(w) + t$ is a nonzero constant times the product of $t + p(r)$ over the five critical points r of p , counted with multiplicity. The squarefreeness of (62) implies that all five critical points are simple and their critical values are distinct.

Remove the five critical values from the target t -line. The inverse image under $t = -p(w)$ is the complex w -line with finitely many points removed, hence is connected. Its degree-six covering therefore has transitive monodromy. A small loop around a simple critical value interchanges exactly two sheets, because the local map has a nonzero quadratic term. The fundamental group of the punctured line is generated by these small loops. Thus its monodromy is a transitive permutation group generated by transpositions.

Place an edge between two sheet labels whenever a generating transposition exchanges them. Transitivity makes this graph connected. Transpositions along a connected graph generate all permutations, so the group is S_6 . Restricting the full target cover to this line cannot enlarge its monodromy. Since the restricted group is already S_6 , the full group is exactly S_6 . $\square$

8.2 A second Morse slice (source 05)

Source 05 proves the same theorem on a different line. Local crossings alone (Corollary 6.7) do not determine the global permutation group; in this example a one-parameter slice supplies all permutations. Set $a = 0$, $b = 1$ and let $d = \tau$ vary. On the target line

$$(c, A, B, D, E) = (1, -1, 0, \tau, 2 - \tau) \quad (63)$$

(that is, $c = 1$, $t = 0$ in (25)) the fiber polynomial is

$$\hat{p}(w) + \tau, \quad \hat{p}(w) = 2w^6 - 2w^5 - w^3 + w. \quad (64)$$

Two exact identities are

$$\hat{p}'(w) = (w - 1)(2w - 1)(6w^3 + 4w^2 + 3w + 1), \quad (65)$$

$$\text{Disc}_w(\hat{p}(w) + \tau) = -4\tau(32\tau + 11)(11664\tau^3 - 6976\tau^2 + 5485\tau - 1372). \quad (66)$$

The last polynomial is squarefree. For instance, its cubic factor has discriminant

$$-3990738242362176 \neq 0,$$

and its values at 0 and $-11/32$ are -1372 and $-9329787/2048$. Hence $\hat{p}$ has five simple critical points with five distinct critical values. (The sign of a critical value is reversed in the τ coordinate, since the equation is $\hat{p}(w) = -\tau$.)

Lemma 8.2 (A connected cover generated by simple branches (source 05)). *A complex polynomial of degree n with $n - 1$ simple critical points and pairwise distinct critical values has monodromy group S_n .*

Proof. This is the argument in the proof of Theorem 8.1, which uses nothing about the degree six. Remove the critical values from the target and their preimages from the source; the result is an unramified degree- n cover whose source, the complex line with finitely many points removed, is connected, so the monodromy is transitive. A small loop around each critical value exchanges exactly two sheets, because the local map at its unique critical point is analytically equivalent to $z \mapsto z^2$, and such loops generate the fundamental group of the punctured target line. The orbits of the group generated by these transpositions are the connected components of their edge graph on the n sheets; transitivity makes the graph connected, and transpositions along the edges of a connected graph generate all of S_n . $\square$

Second proof of Theorem 8.1 (source 05). Away from the five values in (66), the inverse cover of the line (63) is exactly $\hat{p}(w) + \tau = 0$. Lemma 8.2 gives monodromy S_6 on that line. Its image is a subgroup of the monodromy group over $U = \{c \neq 0, H \neq 0\}$, which is in turn a subgroup of the permutations of six sheets; it must therefore be S_6 . Analytic continuation of algebraic root germs preserves every polynomial relation over the base function field, so it acts by automorphisms of the splitting field. Since the resulting permutations already comprise S_6 , the generic splitting-field group, which embeds in S_6 , is exactly S_6 as well. $\square$

Corollary 8.3 (No nontrivial intermediate inverse field (source 05)). *The generic degree-six function-field extension of $\mathcal{F}$ has no proper intermediate field. Consequently $\mathcal{F}$ cannot be written as a composition of two dominant generically finite polynomial self-maps of $\mathbb{A}^5$, each of geometric degree greater than one.*

Proof. By Theorem 3.3, the source function field is generated over the target function field by w . In the S_6 normal closure, the field generated by one root is the fixed field of its point stabilizer S_5 . This is a maximal subgroup of S_6 : any larger subgroup moves the distinguished letter and, using its S_5 subgroup, is transitive on all six letters; its point stabilizer is already S_5 , so its order is $6 \cdot 120 = 720$. Galois correspondence gives the field assertion. A composition with both degrees greater than one would supply a proper intermediate function field, contradicting it. $\square$

This conclusion also says that a generic inverse cannot be described by radicals over the rational function field of the target. The obstruction is not the mere appearance of a sextic: some sextics are solvable by radicals. It is the nonsolvable Galois group just established. The usual solvability criterion and the elementary finite-field cycle method used below are treated in [7, Chapters 3–5].

8.3 An explicit rational certificate

Consider the rational target

$$b_* = (1, 0, 0, 1, 0), \quad f(w) = P_{b_*}(w) = 2w^6 - 2w^5 - w^3 + w^2 + 1.$$

Its discriminant is -993580 . The following factorizations are in the indicated finite fields:

$$f \equiv 2(w^6 - w^5 + w^3 - w^2 - 1) \pmod{3}, \quad (67)$$

$$f \equiv 2(w - 4)(w^5 + 3w^4 - w^3 + 2w^2 + 2w - 5) \pmod{13}, \quad (68)$$

$$f \equiv 2(w + 14)(w - 16)(w - 7)(w - 5)(w^2 + 13w + 14) \pmod{37}. \quad (69)$$

The sextic in (67), the quintic in (68), and the quadratic in (69) are irreducible in their respective fields. The supplied verifier proves this with modular exponentiation and gcds, not by accepting a black-box Galois-group output. Specifically, for a monic polynomial h of degree n over $\mathbb{F}_p$, it checks

$$X^{p^n} \equiv X \pmod{h}, \quad \gcd(h, X^{p^{n/\ell}} - X) = 1 \quad (\ell \text{ prime}, \ell \mid n).$$

These conditions are equivalent to irreducibility, by the factorization of $X^{p^r} - X$ into the monic irreducibles whose degrees divide r .

Theorem 8.4 (Arithmetic S_6 and inverse obstruction). *The polynomial f is irreducible over $\mathbb{Q}$ and has Galois group S_6 . No one of its roots is expressible by radicals over $\mathbb{Q}$. In particular, no full source point of $\mathcal{F}^{-1}(b_*)$ has all five coordinates expressible by radicals over $\mathbb{Q}$.*

Proof. Irreducibility modulo 3 implies irreducibility over $\mathbb{Q}$, and hence a transitive Galois action. At the three displayed primes the reductions are squarefree and the leading coefficient remains a unit. The factorization-cycle theorem gives a 5-cycle from 13 and a transposition from 37 [7,

Theorem 4.28]. To fit the monic integer formulation exactly, apply it to $2^5 f(W/2)$; at these odd primes the change of variable preserves factor degrees.

A transitive subgroup of S_6 containing a 5-cycle has a point stabilizer transitive on the other five labels, and is therefore 2-transitive. Conjugating the transposition gives every transposition. Hence the group is S_6 , in agreement with the standard group criterion [7, Lemma 4.31].

An irreducible polynomial with nonsolvable splitting group cannot have a root expressible by radicals: all conjugates of such a root would belong to a common solvable normal radical extension after adjoining suitable roots of unity, contradicting nonsolvability of S_6 [7, §5, solvability theorem]. Finally $w = \gamma u_1$ is a polynomial in the source coordinates with rational coefficients. If an entire inverse vector had radical coordinates, its corresponding root w would be radical as well. $\square$

This result does not forbid an exact inverse expressed by algebraic numbers or by a root-isolation representation. It forbids a radical-only inverse vector in this explicit fiber. It is therefore directly relevant to the repository's distinction between exact algebraic solving and radical-solvability certificates.

8.4 A second rational target with six nonradical inverse points (source 05)

Source 05 certifies a different target, by different primes and a different group-theoretic deduction. Consider

$$b_{**} = (1, -1, 0, 1, 1), \quad R_{**}(w) = P_{b_{**}}(w) = 2w^6 - 2w^5 - w^3 + w + 1, \quad (70)$$

with $\text{Disc } R_{**} = -1513772 = -4 \cdot 13 \cdot 43 \cdot 677$. It is squarefree, so the fiber contains six complex points; its effective parameter is $(a, b, d) = (0, 1, 1)$, and it lies on the Morse line (63) at $\tau = 1$.

Theorem 8.5 (Arithmetic S_6 certificate (source 05)). *The polynomial R_{**} is irreducible over $\mathbb{Q}$ and has Galois group S_6 . In particular, b_{**} has no rational preimage, and none of its six source points has all coordinates expressible by radicals over $\mathbb{Q}$.*

Proof. The following factorizations hold in the indicated finite fields:

$$R_{**} \equiv 2(w^6 - w^5 + 3w^3 - 3w - 3) \pmod{7}, \quad (71)$$

$$R_{**} \equiv 2(w + 3)(w^5 - 4w^4 + w^3 + 2w^2 + 5w + 2) \pmod{11}, \quad (72)$$

$$R_{**} \equiv 2(w + 76)(w + 98)(w - 132)(w - 114)(w^2 + 71w + 134) \pmod{269}. \quad (73)$$

The degree-six, degree-five and degree-two factors, respectively, are irreducible; the Frobenius tests proving this are listed below, and the supplied script of source 05 executes them with modular polynomial arithmetic. The three primes divide neither the leading coefficient nor the discriminant.

Reduction modulo 7 proves irreducibility over $\mathbb{Q}$, so the Galois group is transitive. For a squarefree reduction at a good prime, Frobenius permutes the roots in cycles whose lengths are the degrees of the irreducible factors; lifting Frobenius to a decomposition group gives an element of the characteristic-zero Galois group with the same cycle type. The reductions therefore supply a 6-cycle, a 5-cycle with one fixed point, and a transposition with four fixed points.

A transitive group of degree six containing a 5-cycle is primitive. Indeed, a nontrivial block system would have blocks of size two or three. An element of order five must fix each block, since there are at most three blocks, and must act trivially within each block, since each has size at most three; this contradicts the presence of the 5-cycle. Finally, a primitive permutation group containing a transposition is the full symmetric group. To see this directly, take all conjugates of that transposition and form their edge graph. The connected components are blocks for the original group, so primitivity forces the graph to be connected, and the edge transpositions then generate S_6 .

A rational source point would yield a rational root $w = \gamma u_1$, contrary to irreducibility. If all coordinates of a source point were expressible by radicals, so would w be, because γ and u_1 are polynomials in the source coordinates with rational coefficients. An irreducible polynomial with Galois group S_6 has no root expressible by radicals: the normal closure of a radical expression has solvable Galois group after adjoining roots of unity, whereas the normal closure of any one root of R_{**} is its S_6 splitting field. $\square$

The Frobenius tests are those displayed before Theorem 8.4. For a monic $f \in \mathbb{F}_p[w]$ of degree n , $w^{p^n} \equiv w \pmod{f}$ together with $\gcd(f, w^{p^{n/\ell}} - w) = 1$ for every prime $\ell \mid n$ is sufficient and necessary for irreducibility: $w^{p^n} - w$ is the product of the distinct monic irreducibles whose degrees divide n , the gcd tests remove factors of every proper divisor degree (every proper divisor of n divides some n/ℓ), and the congruence also forces squarefreeness. For the three nonlinear factors above the complete tests are:

Field	Degree	Tests
$\mathbb{F}_7$	6	$w^{7^6} = w \pmod{f}$; gcds for 7^3 and 7^2 equal 1
$\mathbb{F}_{11}$	5	$w^{11^5} = w \pmod{f}$; gcd for 11 equals 1
$\mathbb{F}_{269}$	2	$w^{269^2} = w \pmod{f}$; gcd for 269 equals 1

All congruences are evaluated by repeated squaring modulo a polynomial of degree at most six; source 05's program does not expand polynomials of degree 7^6 or 11^5 in the ordinary polynomial ring. It also checks each displayed factorization and that the primes are good for the discriminant. The Galois-group certificate is thus small, reproducible, and independent of a black-box command returning a group name.

The two sources' certificates are independent: source 02 uses $b_* = (1, 0, 0, 1, 0)$ with primes 3, 13, 37 and a 2-transitivity argument; source 05 uses b_{**} with primes 7, 11, 269 and a primitivity argument. The conclusions for the omitted point (44) and for b_* , b_{**} are deliberately different. The former target has no geometric preimage. The latter have six geometric preimages, none rational, with a proven radical obstruction. This distinction prevents an arithmetic non-surjectivity statement from being mistaken for a geometric one.

9 Real fibers and arithmetic inversion

Theorem 9.1 (Real cardinalities). *For the real polynomial map $\mathcal{F} : \mathbb{R}^5 \rightarrow \mathbb{R}^5$, the possible fiber cardinalities are exactly $\{0, 1, 2, 3, 4\}$. If $c \neq 0$, the possible values are contained in $\{0, 1, 2, 4\}$. If $c = 0$, the count is 3 for $A^2 + 4B > 0$ and 1 for $A^2 + 4B \leq 0$.*

Proof. For $c \neq 0$ the real source points correspond to the real simple roots of P_b . Suppose all six roots r_i were real, counted with multiplicity. Newton's identities applied to (33) give

$$\sum_i r_i = 1, \quad \sum_i r_i^2 = 1, \quad \sum_i r_i^3 = \frac{5}{2}.$$

But

$$\left| \sum_i r_i^3 \right| \leq \sum_i |r_i|^3 \leq \left(\sum_i r_i^2 \right)^{3/2} = 1,$$

a contradiction. Five simple real roots would force the remaining root to be real as well. Exactly three simple real roots would leave a real cubic factor with no additional simple real root. Its

odd degree forces a real root, and that root must be triple; otherwise a further simple real root remains. This would again make all six roots real. Thus only 0, 1, 2, 4 simple real roots can occur.

For $c = 0$, use $q_b = v^2 + Av - B$ and the one-point section. Two distinct real roots give three points; a double root is deleted, and nonreal roots give no real points. This proves the stated criterion.

All five global values occur. Zero occurs at b_* : write $p(w) = w^2(w-1)(2w^3-1)$. It is nonnegative outside $(2^{-1/3}, 1)$, while on that interval $-1 < p(w) < 0$. Thus $p(w) + 1 > 0$ on all of $\mathbb{R}$. One occurs at the origin, two at $(1, 0, 0, 0, 0)$, and three at $(0, 0, 1, 0, 0)$. Four occurs at $(1, -8, 0, -6, 70)$, whose sextic is

$$2w^6 - 2w^5 - w^3 - 7w^2 + 14w - 6.$$

It has the simple root 1, and sign changes on $(-2, -1)$, $(0, 9/10)$, and $(11/10, 2)$. These give four distinct real roots; the preceding bound proves there are no others. Exact Sturm counts independently verify all five examples. $\square$

Remark 9.2 (All four values occur off $c = 0$ (merge)). The containment for $c \neq 0$ in Theorem 9.1 is an equality: every value in $\{0, 1, 2, 4\}$ is the real fiber cardinality of some target with $c = 1$. Three of them are already realized in the proof above, at $b_* = (1, 0, 0, 1, 0)$ (zero), $(1, 0, 0, 0, 0)$ (two) and $(1, -8, 0, -6, 70)$ (four). For the value one take $b = (1, -1, 0, 0, 1)$. Then $\kappa = 1$, $T = 0$ and $L = 0$ in (1), and

$$P_b(w) = w^3(2w^3 - 2w^2 - 1).$$

The triple root 0 contributes no point. The cubic factor does not vanish at 0 and has discriminant $-140 < 0$, so it has exactly one real root (near 1.2972), and that root is simple. Hence the real fiber has exactly one point. This target is source 05's witness $(a, b, d) = (0, 0, 0)$ for the pattern 31^3 in Proposition 4.6, taken at $c = 1$. For all four targets the merge recomputed the real simple roots of $\text{simp}(q_b)$ exactly with SymPy and mapped each reconstructed real point forward through the expanded map, recovering the target to within 10^{-50} at 80-digit working precision.

A particularly instructive target is therefore b_* . Its geometric fiber has six points, its real fiber is empty, and its rational fiber is empty. These statements coexist without contradiction: geometric surjectivity onto $\mathbb{A}^5 \setminus M$ does not assert real or rational surjectivity.

9.1 An exact inversion procedure over a computable field

The results give a compact algorithm which never expands or eliminates all 2180 monomials of the original map.

- Step 1.** Form κ, T, L and q_b by (1)–(6). For geometric cardinality, compute the gcds in (32) and take the degree, adding one when $c = 0$.
- Step 2.** For inverse points, factor or isolate the roots of $\text{simp}(q_b)$ in the desired field. Over $\mathbb{Q}$, degree-one factors are exactly the rational roots; higher-degree factors represent algebraic inverse points.
- Step 3.** At each simple root evaluate (21), (14), and (22). The only variable denominator is a power of $q'_b(v)$, whose nonvanishing has already been certified.
- Step 4.** If $c = 0$, also solve the triangular equations (15). This supplies the unique point not represented by the $x \neq 0$ chart.

Polynomial degree is at most six throughout the root computation, and drops to two on $c = 0$. The formula handles multiplicities exactly; a near-zero numerical derivative must never be substituted for a gcd test. The accompanying Python module exposes the root polynomial, simple-part calculation, rational reconstruction, and section inverse as reusable functions.

10 Verification, proof boundaries, and formalization

10.1 What the reproducibility package checks

Source 02’s package contains a sparse rational construction of $\mathcal{F}$, a separate verification driver, two coefficient files, and a saved verification log. The driver completed **63 exact checks** under Python 3.13.5 and SymPy 1.14.0. No floating-point arithmetic or network access is used.

The checks include cancellation of the apparent denominators, degree and term counts, the entire $x = 0$ section, the stage inverse, the stage and sweep determinants, the derivative identity, the rescaled root equation, the three normalization relations, the exceptional quadratic, the exact c^2 discriminant factor and its leading coefficient, exclusion of the other all-multiple partitions, every cardinality witness, exact inverse round trips including $\gamma = 0$, the Morse discriminant, finite-field irreducibility certificates, and real Sturm counts.

The saved log is a reproducibility record, not an independent proof assistant certificate. In particular, a finite selection of inverse round trips cannot establish a chart isomorphism. That is why Theorem 3.2 proves the identities on a dense open set and then explicitly proves that they extend over the exceptional locus. Likewise, normality, openness of the embedding, and the monodromy argument are proved mathematically; they are not inferred from a computer-algebra “PASS” message.

10.2 The exact checks of source 05 (source 05)

Source 05’s companion program is independent of source 02’s. It constructs the sweep, rather than entering a precomputed large map. It checks the sweep Jacobian, both compositions of the stage with its inverse, and the stage determinant. It then forms the four quotient polynomials $\hat{S}_i$ and proves the required divisibilities by collecting all coefficients that could obstruct them; the boundary jets and their triangular Jacobian are checked exactly. For the fiber analysis it checks the sextic elimination, identity (19), and all four sweep equations modulo P after reconstruction. It verifies the square polynomial and its discriminant, the sample factorization table, the triple-root cofactor and its discriminant, the omitted-surface parametrization, and the c -adic discriminant coefficient. Finally it checks the critical-value polynomial of the second Morse slice and the three modular factorizations, with irreducibility tests performed by repeated squaring and gcd computations. The recorded run (Python 3.13.5, SymPy 1.14.0) reports twelve PASS groups, then **ALL CHECKS PASSED**, followed by two lines restating the proof boundary.

All these are rational or finite-field calculations. No floating-point roots, random samples, heuristic factor selections or remote computations enter the certificate. The full coefficient expansion of Δ is regenerated by the program. The script is an independent computational audit of explicit identities, not an alternate proof of every theorem: it cannot replace the proof that the seven multiplicity partitions are exhaustive, the local analytic coordinate argument, the properness argument, or the group-theoretic deductions. These are given in the text.

Neither suite checks the trace-form statement of Theorem 6.1 ($\text{Disc}(\mathcal{B}) = H/256$), nor source 05’s local normal forms beyond the quartic discriminant (61); the intake’s independent spot check (not shipped) confirmed the trace-form identity at three rational targets. In preparing this merge, the renamed formulas of source 05 printed here (the quotient polynomials of Appendix C, identity (12), the restriction (29), the triple-root family, the second Morse slice, and the factorizations modulo 7, 11, 269) were re-checked with SymPy 1.14.0 against the construction (9)–(11). That check is not shipped.

10.3 A suitable Lean development boundary

A formalization should begin with the small algebra, not with a giant five-variable graph ideal. The natural dependency order is:

- (1) Prove the elimination and derivative identities in a polynomial ring. These are finite ring equalities with rational coefficients.
- (2) Formalize the multiplicity-one factor construction (32), including the zero-characteristic hypotheses and constant-polynomial edge case. This yields a reusable exact fiber-count checker.
- (3) Establish the six-element algebra and its multiplication rules. Prove spanning by closure under the two generators and independence by the degree-six field relation. Then derive the discriminant basis change.
- (4) Formalize the source chart and its exceptional section, either as localized coordinate-ring isomorphisms or first as field-valued point bijections with explicit nonvanishing hypotheses.
- (5) Add the geometric layer: normality of the three charts, the open immersion criterion, and the branch/nonproperness interpretation. This stage needs substantially more algebraic-geometric infrastructure than the finite polynomial certificates.

No Lean source is supplied as if it were checked. A future kernel audit should report separately which finite identities, pointwise statements, and scheme-theoretic conclusions have actually been formalized.

10.4 A narrow, layered Lean route (source 05)

Source 05 proposes a parallel plan, also explicitly a proposed implementation plan and not a report of completed Lean work. Its second layer corresponds to steps (1) and (4) above, its third to the exceptional section of step (4), its fourth to step (2) together with the Galois certificates, and its fifth to step (5). Its first layer (the multiplicity patterns) has no counterpart above, and it has no counterpart of step (3). The repository's style of connecting exact algebraic certificates to mathematical semantics suggests the following order.

First layer: the sextic family. Define $P = 2w^6 - 2w^5 - w^3 + aw^2 + bw + d$ as a univariate polynomial over a characteristic-zero field. Prove the coefficient identities and the power-sum constraints. Formalize the two-root contradiction (Lemma 4.4) separately from any multivariate geometry. The all-multiple-root theorem can then be proved by multiplicity partitions and coefficient comparison. This yields the essential omitted-parameter result in a small algebraic development.

Second layer: reconstruction. Represent the stage and the sweep as multivariate polynomials, and use normalized polynomial certificates for (19). State the reconstruction theorem first for points with explicit nonzero hypotheses $c \neq 0$ and $P'_b(w) \neq 0$. Avoid a definition that treats field division at zero as if it were a removable quotient. The polynomial extension (Proposition 2.1) must be built from divisibility, with the quotient identities proved before evaluation.

Third layer: the hyperplane. The triangular map (15) and the equation $x^2(A^2 + 4B) = 1$ require no scheme theory. They can provide a complete set-theoretic fiber-count theorem, including the exceptional locus, before any formalization of properness or analytic monodromy is attempted.

Fourth layer: finite algebra and certificates. Formalize the simple-root-factor formula and its interpretation. The finite-field irreducibility checks can be encoded as polynomial remainder and gcd certificates. Connecting the cycle types to S_6 requires an explicit Frobenius bridge and elementary permutation-group lemmas; it should not be hidden behind an unchecked external Galois-group command.

Fifth layer: geometry and topology. Scheme localization, the normalization of $\mathcal{D}$, analytic preparation, and the covering-space argument are substantial separate interfaces. A formalized algebraic fiber theorem should not be advertised as a formalized nonproperness or monodromy theorem until those interfaces are completed.

This layered plan gives useful, independently reviewable endpoints. It also keeps the small exact certificates outside the trusted boundary until a proved checker or a direct kernel proof verifies their semantic meaning. None of the layers exists in ProveIt at the time of this report (Section 1.2).

10.5 Why the argument is not a numerical discovery claim

The important transitions are algebraic. A root is deleted because an exact derivative is zero. An entire fiber is empty because an exhaustive list of multiplicity partitions reduces to one square. The finite model is normal because its charts are smooth, not because sample fibers look correct. The Galois group is S_6 because of exact permutations forced by finite-field factorizations and a connected Morse cover. These arguments continue to hold at targets which numerical root solvers would find highly ill-conditioned.

11 Further research questions and proposed projects

The following questions are specific continuations of the results above. They are proposed directions, not additional theorems established here.

11.1 Complete the adjacent higher-dimensional fiber problems

1. The degree-twelve map F_7 . Gao supplies a univariate fiber polynomial of degree twelve [3, §4.6]. Determine whether its exceptional degeneration is resolved by successive versions of Theorem 7.1. A concrete first deliverable is an integral basis and a factorization of its normalized discriminant, followed by an exhaustive empty-fiber classification. The F_6 analysis suggests that the exceptional hyperplane must be treated independently of the generic elimination equation.

2. The maps F_4 and F_5 . Their published quintic tangency and degree-ten resultant descriptions leave related fiber problems [3, Appendix A.3]. Derive a derivative or subresultant criterion that distinguishes an actual source point from a deleted tangency parameter. The main risk is that a resultant introduces extraneous solutions; each proposed root representation should be accompanied by an explicit inverse chart.

11.2 Refine the geometry of the sextic example

3. Singular strata of H . The simple-root gcd criterion gives the correct count at every point, but does not constitute a Whitney stratification. Determine the singular locus of the discriminant via (55), separating triple roots, two double roots, and higher coincidences. Describe their intersections with $c = 0$ and the omitted surface M . An attractive deliverable is a small set of parametrizations or subresultant ideals rather than a large expanded elimination basis.

Status after the merge. Partly answered by source 05, and re-scoped. Theorem 6.6 gives the local analytic type of $\mathcal{D}$ at every point of coefficient space and the singular-locus criterion (a root of multiplicity at least three, or at least two distinct repeated roots); Proposition 6.2 gives the normalization of $\mathcal{D}$; Corollary 6.7 describes $V(H)$ along M ; and Theorem 6.4 gives its behaviour along $c = 0$. What remains open is a Whitney stratification, explicit parametrizations or ideals of the singular strata of $V(H)$ in the five-dimensional target, and their full intersection pattern with $c = 0$. Source 05 likewise claims no Whitney stratification.

4. Quantitative escape rates. For each deleted component E_i , construct explicit Laurent-series source arcs whose images tend to a prescribed point of $c = 0$. Compute the leading pole orders of all five source coordinates. Repeat near a generic ramification point and near the triple-double locus M . This would distinguish the unramified and ramified mechanisms dynamically, not just set-theoretically.

5. The smooth affine normalization Z . Determine its divisor class group, Picard group, and algebraic automorphisms. Is the explicit six-generator order monogenic after a target automorphism or after a smaller localization? Any answer must respect the quadratic-plus-quartic special fiber (48); the naive power basis cannot do so.

11.3 Extend the algebraic and arithmetic methods

6. Higher contractions. Replace $w^2J - cTw - c^2L$ by a family whose leading degeneration contains w^r , with $r \geq 3$. Find hypotheses under which $r - 1$ explicit divided basis elements suffice to normalize it. The desired theorem should state a finite-free basis, multiplication rules, discriminant correction, and checkable normality conditions.

7. Arithmetic image and solvable fibers. Classify rational parameter subvarieties on which the sextic Galois group drops to a solvable subgroup. Distinguish those with a rational source point from those with only nonrational radical inverse points. The explicit S_6 certificate is a starting obstruction, not a classification of all specializations. This problem connects directly to ProveIt's radical-solvability and exact polynomial-solving projects.

8. Good reduction. Analyze the root chart and normalization in positive characteristic, separating primes excluded by the map's coefficients or determinant from primes affecting discriminants such as 401. The characteristic-zero proof must not simply be reduced modulo an arbitrary prime: derivative multiplicities, separability, and the stage determinant require independent checks.

11.4 Formal and structural questions

9. A certified root-chart toolkit. Build a reusable library that takes a polynomial identity for a graph, a reconstruction tuple, and nonvanishing certificates, and returns a checked fiber bijection. Add support for exceptional source components instead of silently localizing them away. This would serve the Jacobian, radical-expression, and algebraic-number parts of ProveIt simultaneously.

10. Rigidity versus finite-cover complexity. Gao's middle-branch rigidity questions concern when a higher-dimensional construction is only a suspension of a lower-dimensional one [3, §4.5.2]. Investigate whether normalized-cover invariants, especially monodromy and the number of unramified boundary components, obstruct such suspensions. The present S_6 computation alone is not a proof of a new rigidity theorem; a comparison theorem for finite normalizations would be needed. Source 05's Corollary 4.10 excludes a suspension of F_6 from dimension at most three, by the geometry of the omitted surface; direction 13 below asks about dimension four.

11.5 Directions from source 05 (source 05)

Source 05 lists eight further questions, numbered 11–18 here. The completed calculations make them accessible; they are not assertions that the corresponding results already hold. Where source 02’s results answer or overlap one, this is stated.

11. A global finite completion across $c = 0$ (answered by source 02). Over $c \neq 0$, (26) gives a smooth finite completion with a transparent deleted ramification divisor. Source 05 asked for an explicit finite normalization over the whole target $\mathbb{A}^5$, with the original affine source located inside it; in particular, for the boundary components above $c = 0$, their degrees, and how the three lost sheets interact with the two-sheet discriminant locus at (59). Theorems 5.1 and 5.2 answer this: $Z = \text{Spec } \mathcal{N}$ is finite free of rank six and smooth; the source is $Z \setminus (\text{Ram}(\pi) \cup E_1 \cup E_2 \cup E_3)$; each $E_i \simeq \mathbb{A}^4$ maps isomorphically (degree one) onto $c = 0$, is unramified and is disjoint from $\text{Ram}(\pi)$; and by (48) the two remaining sheets over $c = 0$ form the quadratic factor $\zeta^2 + A\zeta - B$, which ramifies exactly along $A^2 + 4B = 0$. Refinements, such as escape rates along each component, are source 02’s question 4.

12. Higher-codimension omitted sets. The omitted surface arises from three independent double roots of a degree-six fiber polynomial. Can a tangent-sweep construction realize r independent double roots, with an omitted locus of codimension r , for arbitrary r ? Within a local model consisting of disjoint simple folds and no surviving sheets, degree at least $2r$ is necessary simply by counting the two sheets of each fold; the present example realizes equality for $r = 3$. A theorem for general Keller maps would require much more than this local counting observation. A practical route is to start with a family of degree- $2r$ polynomials having a unique square member and an r -dimensional transversal coefficient slice, then solve the stage divisibilities that turn its ramification into escape. The difficult part is not the square polynomial, but constructing a global polynomial source with constant Jacobian.

13. The remaining four-dimensional stabilization question. Corollary 4.10 excludes every stabilization from dimension at most three. Is F_6 nevertheless equivalent to $\Phi \times \text{id}_{\mathbb{A}^1}$ for a four-dimensional Keller map Φ of geometric degree six? Its omitted curve would have to have a cylinder isomorphic to $\mathbb{G}_m \times \mathbb{A}^1$, so the omitted surface is compatible with such a possibility rather than excluding it. The embedding of that surface, the normalization of the escape divisor, and the distribution of boundary sheets in a global finite completion may decide this remaining case. (That completion is now Z ; see direction 11.) The composition obstruction of Corollary 8.3 does not settle it: adding an identity coordinate does not factor the generic degree into smaller degrees.

14. The degree-twelve fiber polynomial of F_7 (overlaps question 1). Apply the derivative-localization method to the other five-dimensional construction in [3]. A useful first target is not the entire discriminant expansion, but a characterization of all fiber polynomials without simple roots. Degree twelve admits many more multiplicity partitions, so the first questions are which Newton-sum constraints survive and whether a small effective parameter space remains transverse to every allowed contact pattern.

15. Real image and real fiber chambers (re-scoped). The reconstruction formulas are defined over $\mathbb{Q}$, so for real targets with $c \neq 0$ the real fiber is exactly the set of real simple roots of P_b ; on $c = 0$ the real count is three when $A^2 + 4B > 0$ and one otherwise. Theorem 9.1 of source 02 has since determined the possible real fiber cardinalities, $\{0, 1, 2, 3, 4\}$, and the values off $c = 0$ are exactly $\{0, 1, 2, 4\}$, each attained at a target with $c = 1$ (Remark 9.2). What remains open is source 05’s actual request: the connected real chambers, the real fiber count on each, and the semialgebraic boundary of the real image. Sturm sequences for the exact simple-root factor provide an immediate computational route, but a useful result should describe the chambers geometrically rather than merely output a quantifier-elimination formula.

16. Arithmetic specialization and positive characteristic (overlaps questions 7 and 8).

For rational targets, determine the distribution of specialized Galois groups, the arithmetic of one-point and two-point fibers, and the loci admitting rational source points. The explicit S_6 targets give starting certificates, not such a distribution theorem. The construction also invites reduction modulo primes, but characteristic-zero conclusions must not be transferred indiscriminately. The denominator 5, the determinant -290 , the cubic discriminant $-401/128$, and the integer moment obstructions already reveal several exceptional primes. Determine exactly which fiber patterns and omitted-locus formulas survive at each prime, and separate inseparability from genuine changes in the coefficient constraints.

17. The middle-branch rigidity problem remains separate (overlaps question 10). Gao's Problem 4.8 concerns whether the entire middle branch of a mixed direction-field construction is a suspension. The present results analyze a bottom-branch example and do not answer that question. A productive next step is to derive a fiber-polynomial or incidence invariant for the middle branch and test whether the suspension restriction imposes a detectable factorization on its monodromy or ramification geometry. Such a test could produce either a rigidity proof or a systematically searchable counterexample class.

18. Formal, reusable fiber-classification infrastructure (overlaps question 9). A reusable library should take a monic polynomial family and an explicit localization condition, compute the simple-root factor, and connect it to the fibers of a polynomial map via a proved reconstruction equivalence. The present case is small enough to serve as a complete benchmark: it includes generic fibers, repeated roots, an exceptional hyperplane, an omitted locus, and a finite-field Galois certificate. Formalizing this example would be more valuable than checking isolated collisions, because it would verify the transition from elimination algebra to a global classification.

12 Conclusion

The central open task selected from the repository's surrounding research program was not the existence of another collision, but the exact behavior of every fiber of an already explicit high-degree map. The derivative identity makes that task accessible. The uniform chart then exposes the exceptional quadratic, and one divided basis element repairs the finite model. These steps determine the image, all geometric and real cardinalities, the branch and nonproperness loci, and a concrete radical obstruction.

The resulting model is small: a sextic, a quartic, a quadratic exceptional fiber, and three multiplication rules. It retains the geometry hidden by the large expanded map and provides a practical target for independent review, further generalization, and formal verification.

Source 05 draws the same conclusion from the other side (source 05). The central fact is not simply that a sextic appears in an inverse problem: its derivative is exactly the multiplier that decides which inverse points remain in affine space. Once this identity is proved, the question of omitted targets becomes a question about a rigid coefficient slice with no simple roots, and Newton sums reduce it to a single square of a cubic. The exceptional hyperplane contributes no omitted points and is controlled by one quadratic equation. The resulting picture is exact: a smooth codimension-three omitted surface, all six possible fiber sizes and all seven multiplicity patterns, a two-component nonproperness divisor, the normalization and all local analytic types of the discriminant, a dimension-three stabilization obstruction, and full symmetric monodromy. These are concrete extensions of the polynomial-map research direction represented in ProveIt. Their algebraic core is accompanied by exact certificates; the remaining formalization and research problems are separated from the proved statements.

A Reproducing the calculations

Both sources' verification suites are shipped, byte-identical to the delivery but renamed with the prefixes `02-six-sheets-` and `05-keller-fibers-`. Each script writes its outputs *beside itself under its delivered, unprefixed names*, and source 02's verifier imports `build_map` by its delivered module name. Rerun them therefore only on a copy laid out as delivered, never in the report directory; this keeps the shipped records intact.

Source 02 (SymPy 1.14.0, pinned in `data/02-six-sheets-requirements.txt`). In a scratch directory with subdirectories `code` and `data`, copy `code/02-six-sheets-build_map.py` to `code/build_map.py` and `code/02-six-sheets-verify.py` to `code/verify.py`, then run

```
python -m pip install sympy==1.14.0
python code/verify.py
```

from the scratch directory. It rewrites `data/map_coefficients.json`, `data/normalized_discriminant.json` and `data/verification_report.txt` there; compare them with the shipped `data/02-six-sheets-*` files. Run standalone, `build_map.py` also writes `data/zero_section.txt`, which is not shipped.

Source 05 (same pin, `data/05-keller-fibers-requirements.txt`). Copy `code/05-keller-fibers-verify.py` to `verify.py` in a scratch directory and run

```
python verify.py > verification.txt
```

It writes `sextic_discriminant.txt` beside itself even without redirection; compare the two outputs with `data/05-keller-fibers-verification.txt` and `data/05-keller-fibers-sextic_discriminant.txt`. The function `complex_fiber_size`, imported as `fromverifyimportcomplex_fiber_size`, accepts an exact rational target (c, A, B, D, E) and counts geometric points by the simple-root factor and (28); it is not a count of rational or real preimages.

On Windows, text-mode writes add carriage returns; the shipped files have none, and the intake's reruns matched them byte for byte after stripping carriage returns. Neither suite needs network access after installation, and neither needs a L^AT_EX installation. This report compiles independently of Python with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

Shipped file (delivered name)	Purpose
code/02-six-sheets-build_map.py (code/build_map.py)	Source 02: sparse rational construction and exact denominator cancellation.
code/02-six-sheets-verify.py (code/verify.py)	Source 02: the 63-check certificate and reusable rational inverse functions.
data/02-six-sheets-map_coefficients.json	All five polynomial coordinates; monomial exponent order is (x, y, z_1, z_2, z_3) .
data/02-six-sheets-normalized_discriminant.json	Sparse H in coordinate order (c, A, B, D, κ) ; $\kappa = E + B^2 - AD$ (written K in the file).
data/02-six-sheets-verification_report.txt	Saved output of source 02's completed exact run.
data/02-six-sheets-requirements.txt	sympy==1.14.0.
02-six-sheets-SOURCE_AUDIT.md	Source 02: repository revision, source scope, and novelty boundary.
code/05-keller-fibers-verify.py (verify.py)	Source 05: exact rational and finite-field checks, and <code>complex_fiber_size</code> .
data/05-keller-fibers-verification.txt	Saved output of source 05's run (twelve PASS groups).
data/05-keller-fibers-sextic_discriminant.txt	The expanded 54-term $\Delta(a, b, d)$, written in source 05's coefficient names (a, b, c) , so its c is d here.
data/05-keller-fibers-requirements.txt	sympy==1.14.0.
05-keller-fibers-SOURCE_AUDIT.md	Source 05: pinned sources, comparison scope, and attribution.

Every coefficient in source 02's JSON files is a rational string; no precision parameter is needed. Its verifier regenerates the coefficient files from the formulas rather than accepting them as trusted input. A failure raises an exception and prevents the final all-checks-passed message. There are no random checks in either verifier.

B Source audit and attribution

The repository reference observed during the audit was

e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9.

The main directly inspected project files were

Algebra/JacobianConjecture/README.md
Algebra/JacobianConjecture/Research/README.md.

Their returned blob hashes were respectively

67e25dcb94cb7a38fb508ea65e3a35455df859ac
f0605a788b98bed23df91318aa2069e6802fc66e.

The root README and a walkthrough search result were used for orientation. The root README's general formalization claims were not independently rebuilt in this work, and none is used as a substitute for the specific algebraic proofs above.

The main external mathematical source was Gao's version 1, dated 31 July 2026, particularly §4.5.1, Theorem 4.5, the concluding remarks, and Appendix A.3. Its definition of F_6 , determinant, generic degree, and elimination polynomial are explicitly credited. The derivative identity, uniform chart, complete cardinality criterion, exact omitted surface, six-element normalization, normalized

discriminant and boundary, and the geometric and arithmetic S_6 proofs are the extensions developed in this article. This describes their relation to the inspected sources, not a guarantee that no equivalent result exists elsewhere.

The primary literature search checked the cited paper's available version and searches combining its identifier or author with F_6 , fiber, normalization, and Jacobian terminology. No directly matching extension was identified in that bounded search. It was not an exhaustive review of all unpublished work, repository branches, or later inaccessible material. The Stacks Project and Milne are cited for the standard geometric and Galois-theoretic facts used in the proofs.

Source 05's audit (source 05). Source 05 inspected the same pinned commit. Its principal repository sources were `Algebra/JacobianConjecture/Lean/JacobianConjecture/Counterexample.lean`, `Algebra/JacobianConjecture/Research/README.md`, and `ProveIt_Walkthrough.tex`, specifically the walkthrough's reference to Gao's account as external context for the repository's formal determinant and collision identities [2]. It also inspected the Fabius overview during topic selection, which is not a mathematical dependency. It was a targeted research survey, not a full audit of every repository file or a build of the Lean development; no repository file was modified or uploaded. From Gao's version 1 it used §4.5.1 (the definition of F_6 , its stage and sweep, determinant, and sextic fiber polynomial) and Appendix A.3 (the deferred fiber stratifications); it reports inspecting the rendered formulas on PDF pages 22 and 23 and the deferred statement on page 31. (Source 02 cites printed pages 23 and 24; the two are consistent with an offset between PDF and printed page numbers, which this merge did not check against the paper.) Source 05 credits the map to Gao, and claims as new the derivative-localization identity, the exhaustive multiplicity classification, the complete hyperplane fibers, the unique square member, the explicit omitted surface, the stabilization obstruction from dimension three, the global nonproperness equation and c -adic coefficient, the normalization and local analytic types of the discriminant, full S_6 monodromy, the intermediate-field obstruction, and its rational S_6 target. Its targeted searches combined Gao, F_6 , fiber, monodromy, omitted surface, codimension three, and the distinctive rational coefficients of the omitted parameter; no overlapping complete treatment was located. This limited search is not a comprehensive priority check and does not rule out independent or newly posted work. Source 05 credits only the construction to Gao; source 02 also credits the degree profile and the four-point fibers over the nonzero axis. Neither attribution was checked against Gao's paper by this merge.

C The four sweep quotients (source 05)

Substituting $(w_1, w_2, w_3) = (\gamma u_1, \gamma^3 u_2, \gamma^4 u_3)$ into the sweep coordinates S_i of Section 2 and dividing by γ^i gives

$$\hat{S}_1 = 1 - 2u_1 + 5\gamma u_1^2 + \gamma^2(-2u_1^3 + 2u_2) + \gamma^3(8u_1^4 - u_3) + \gamma^4(-10u_1^5 + 2u_1 u_3), \quad (74)$$

$$\hat{S}_2 = u_1 - u_1^2 + \gamma(4u_1^3 + u_2) + \gamma^2(-2u_1^4 + 2u_1 u_2) + 8\gamma^3 u_1^5 + \gamma^4(-10u_1^6 + 2u_1^2 u_3), \quad (75)$$

$$\begin{aligned} \hat{S}_3 = & u_1^2 + u_2 + \gamma(3u_1^4 + 2u_1 u_2) + \gamma^2(-2u_1^5 + 2u_1^2 u_2) \\ & + \gamma^3(8u_1^6 + u_1^2 u_3) + \gamma^4(-10u_1^7 + 2u_1^3 u_3), \end{aligned} \quad (76)$$

$$\begin{aligned} \hat{S}_4 = & u_1^4 + (1 - 2u_1)u_2 + u_3 + 3\gamma u_1^2 u_2 + \gamma^2(-u_1^6 + u_2^2) \\ & + \gamma^3(8u_1^4 u_2 - 2u_1^3 u_3 - u_2 u_3) + \gamma^4(-10u_1^5 u_2 + 2u_1^4 u_3) - \gamma^6 u_1^2 u_3^2. \end{aligned} \quad (77)$$

These expressions have 9, 9, 10 and 13 expanded monomials before the stage substitution. They suffice to reproduce the entire map, the divisibility proof of Section 2, and the exceptional-hyperplane calculation of Section 3.4, without printing the large final coordinate expansions. (Source 05 abbreviates (u_1, u_2, u_3) as (u, v, w) in this appendix only; this report does not, since v and w are the chart and root variables here.)

D Provenance of this merged report

This report was built in the write phase of ProveIt’s incoming batch 36 from two manuscripts of that batch, both AI-assisted (prepared with ChatGPT), both dated 24 September 2026, and both pinned to ProveIt commit `e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9`. They were placed in commit `1a1396d4d`.

- *Source 02* (batch 36 manuscript 02; archive `ProveIt_Six_Sheets_Research`; main file `Six_Sheets_Three_Escapes.tex`, 22-page PDF): the base. It contributes the text of Sections 1–12 and Appendices A–B apart from the tagged insertions: the uniform chart across $c = 0$, the explicit finite normalization and its boundary, the normalized discriminant and trace form, the nonproperness locus, the general quadratic-contraction lemma, geometric and arithmetic S_6 (primes 3, 13, 37), the real fiber sizes, the inversion procedure, its Lean plan and questions 1–10.
- *Source 05* (batch 36 manuscript 05; archive `ProveIt_Keller_Fibers_Research`; main file `article.tex`, 24-page PDF): merged in. It contributes the material tagged (source 05): the multiplicity patterns and the three-distinct-roots lemma, the reconstruction over $c \neq 0$ and the finite completion, the hyperplane computed on the source, the stabilization obstruction, the normalization of $\mathcal{D}$, transversality along $c = 0$, the local analytic types, normal crossings and $(\mathbb{Z}/2)^3$ along M , a second Morse slice, the intermediate-field corollary, a second arithmetic certificate (primes 7, 11, 269), its exact checks, its layered Lean plan and directions 11–18.

Choices made in the merge.

- (1) Source 02 is the base: its chart is a scheme isomorphism across the whole target, its normalization lemma holds over any domain, and its finite normalization answers source 05’s question on a global completion (direction 11).
- (2) Source 05 is printed in source 02’s notation (Table 1); no normalization changed. Renamed symbols of source 05: $(C, p_1, \dots, p_4) \rightarrow (c, A, B, D, E)$; $\xi_i \rightarrow a_i$; $(t, s, r) \rightarrow (w_1, w_2, w_3)$; $E_i \rightarrow \hat{S}_i$; $J_i \rightarrow \beta_i$ (the section coordinates); $R_{a,b,c}(T) \rightarrow P$ with root w and coefficients (a, b, d) ; $D \rightarrow \Delta$; $\tilde{D} \rightarrow H$; $H \rightarrow A^2 + 4B$; $\eta \rightarrow t$; $\mathcal{M} \rightarrow M$; $S_F \rightarrow \text{NP}(\mathcal{F})$; $Q \rightarrow Q_*$; $H_4 \rightarrow h_4$; $K \rightarrow \Lambda$; power sums $p_j \rightarrow N_j$; $\delta \rightarrow \theta$ (two roots); (u, v, w) , $Z \rightarrow (\sigma_1, \sigma_2, \sigma_3)$, ω (local types); $Z_i \rightarrow \omega_i$ (root clusters); k (count of vanishing ϵ_i) $\rightarrow j$; $P(T)$, $c \rightarrow \hat{p}(w)$, τ (Morse slice); $q_* \rightarrow b_{**}$; $N \rightarrow \Omega_\Phi$; the lifted factors $A_C, B_C \rightarrow \phi_c, \psi_c$; $d, e, f \rightarrow e_1, e_2, e_3$ (the cubic’s coefficients); $d \rightarrow d_1$ (the gcd in the fiber count); $k[s, t] \rightarrow k[y_1, y_2]$; the Morse-lemma degree $d \rightarrow n$; and, in Appendix C, $(u, v, w) \rightarrow (u_1, u_2, u_3)$.
- (3) One symbol of source 02 was renamed: its nonproperness set $\mathcal{N}_{\mathcal{F}}$, which clashed with its normalization algebra $\mathcal{N} = \mathcal{N}$, is $\text{NP}(\mathcal{F})$ here.
- (4) Source 02’s typographical defect in (33), where “\text” had been corrupted into a tab followed by “ext” (its PDF printed “extterms of degree at most two”), is corrected.
- (5) Results proved by both sources are printed once and credited to both: polynomiality and $\det J = -290$, the section at $x = 0$, the derivative identity, the reconstruction, the gcd count, the spectrum, the unique square, M , $\Delta = c^2 H$ with (53), the irreducibility of Δ and H , the nonproperness locus, geometric S_6 and the radical obstruction. Genuinely different proofs of source 05 are kept as marked second routes: by finite jets (polynomiality), by explicit expansion (derivative identity), on the source (hyperplane count), through the effective parameters (omitted surface), through Newton sums (unique square), by coprime factor lifting (c -adic factor), through the finite completion (nonproperness), and on a different Morse line (geometric S_6). Source 05’s arithmetic certificate is a different target and is printed as its own theorem.

- (6) Source 05’s corollary on the rational omitted point is printed as a remark, since the point is already the zero-point witness of Table 2. Its Lemma on connected covers is printed as the standalone Lemma 8.2, beside its second proof of Theorem 8.1; its proof notes that source 02’s proof of that theorem contains the same argument. Its appendix of Frobenius tests is printed in Section 8.4.
- (7) Source 05’s eight directions are printed as 11–18. Direction 11 (a global finite completion) is answered by source 02 and marked so; direction 15 (real chambers) is re-scoped after source 02’s Theorem 9.1, and the merge checked that all four values 0, 1, 2, 4 occur off $c = 0$ (Remark 9.2), a point neither source raised; source 02’s question 3 is re-scoped after source 05’s local types. No other question is answered by the other source.
- (8) Source 02’s source-audit appendix and source 05’s audit are kept as historical records of the pinned commit; the files they inspected were unchanged from the pin to the placement commit. The reproducibility appendix was rewritten for the shipped file names.

Label prefixes: every label carries `f6:.` Labels of source 02 keep their names after the prefix; labels introduced for source 05’s material carry `f6:kf:.`; the six other labels added by the merge (five on editorial text, one on source 02’s previously unlabelled Conclusion) (`f6:sec:merge`, `f6:sec:dictionary`, `f6:tab:dictionary`, `f6:rem:realoffc0`, `f6:sec:conclusion` and `f6:app:provenance`) carry `f6:` only.

References

- [1] Vladimir Reshetnikov, *ProveIt*, GitHub repository, Jacobian-conjecture project and research README; revision `e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9`. <https://github.com/VladimirReshetnikov/ProveIt>. Source 05 cites the same revision, in particular the Lean module `Counterexample.lean` and the research overview in `Algebra/JacobianConjecture`.
- [2] Vladimir Reshetnikov, *ProveIt Walkthrough*, source `ProveIt_Walkthrough.tex` at the same revision; its Jacobian discussion cites Gao’s account as external context, separate from its local formal identities (cited by source 05). https://github.com/VladimirReshetnikov/ProveIt/blob/e21766d04c2b8a9b2cdba0cd43e563024b3bd1b9/ProveIt_Walkthrough.tex.
- [3] Shuhong Gao, *Counterexamples to the Jacobian conjecture in dimensions greater than two*, arXiv:2608.00222v1, 31 July 2026. <https://arxiv.org/abs/2608.00222v1>. In particular §4.5.1, Theorem 4.5, and Appendix A.3.
- [4] The Stacks Project Authors, *The Stacks Project*, Tag 02GH, “Étale morphisms,” including the lemma that a morphism between schemes étale over a common base is étale. <https://stacks.math.columbia.edu/tag/02GH>.
- [5] The Stacks Project Authors, *The Stacks Project*, Tag 025G, Theorem 41.14.1: universally injective étale morphisms are open immersions. <https://stacks.math.columbia.edu/tag/025G>.
- [6] The Stacks Project Authors, *The Stacks Project*, Tag 0BJF, Lemma 49.3.1: the discriminant of a finite locally free morphism and its étale locus. <https://stacks.math.columbia.edu/tag/0BJF>.
- [7] James S. Milne, *Fields and Galois Theory*, version 5.10, September 2022, 144 pp. Theorem 4.28, Lemma 4.31, and Chapter 5. <https://www.jmilne.org/math/>.